INTEGRAL TOPOLOGY AND SMALL EIGENVALUES IN CALABI–YAU CONIFOLD TRANSITIONS

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Abstract. We determine the leading small spectrum of the Hodge Laplacian on real two-forms for Ricci-flat metrics in polarized Calabi–Yau conifold transitions. We assume a simply connected projective small resolution and a fixed smoothing that smooths every node to first order, approached along a fixed ray. Exactly as many positive eigenvalues tend to zero as there are divisor directions lost under smoothing, and their decay is inverse-logarithmic in the smoothing parameter. The leading coefficients are 2π times the generalized eigenvalues of the squared exceptional-period form relative to the harmonic norm after projection off all surviving cohomology. We compute this norm from global intersections. The divisor-period restriction map also determines the integral relations among cohomological vanishing cycles. The small eigenspaces are coexact and primitive of real type $(1,1)$, with a uniform gap in the remaining positive two-form spectrum. Applications to four-node and eight-node toric transitions give three and two distinct coefficients, respectively. For the sixteen-node split quintic with Kähler class $c_1(\mathcal{O}(1))$, the coefficient is 10π; we also calculate its integral rings, attachment and specialization maps, primitive vanishing lattice and monodromy image of index sixteen.

1. Introduction

We determine the leading small positive two-form eigenvalues of the Ricci-flat metrics in a polarized Calabi–Yau conifold transition. Their number is the loss of second cohomology under smoothing. Their coefficients depend on the periods of divisor classes over exceptional curves and on the global polarized intersection form. The same period map determines the integral relations among the vanishing cycles. Thus the topological and spectral conclusions use the same marked geometric transition. Local node types alone do not determine either the compact cohomology ring or these spectral coefficients.

We calculate the global data for the split quintic and for two toric transitions constructed in earlier work. The two toric applications use established geometry from the stability–degeneration series and from the separate Calabi–Yau smoothing series. We denote Papers IV–VI of the latter series by CY IV–VI below; these are distinct from the numbered stability–degeneration papers. The spectral theorem and its proof come first, followed by the compact applications and the precise roles of their geometric sources.

1.1. **Geometric hypotheses and the main theorem.** Let X_0 be a normal projective threefold with trivial dualizing sheaf, with exactly $k \geq 1$ ordinary double points and no other singularities. Let

$$q : Y \longrightarrow X_0, \qquad f : \mathcal{X} \longrightarrow \Delta$$

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be respectively a projective small crepant resolution and a proper flat polarized smoothing over a complex disc. Assume that Y is smooth and simply connected, that the relative canonical bundle of f is trivial, and that $X_t = f^{-1}(t)$ is smooth for $0 < |t| < t_0$. At each node the local smoothing function τ_i is required to satisfy

$$\tau_i(0) = 0, \qquad \tau_i'(0) \neq 0. \tag{1}$$

This is a condition on a genuine family over a disc. Let $C_i \simeq \mathbb{P}^1$ be the exceptional curves and impose

$$b_2(Y) = 2, \qquad b_2(X_t) = 1. \tag{2}$$

The polarization on X_0, its pullback to Y, and its extension to X_t are denoted by H. It is an integral class, with no implicit factor of 2π. Choose an integral divisor class D independent of H over $\mathbb{R}$, and put

$$v = H^3, \qquad a = DH^2, \qquad d = D^2H, \qquad b_i = D \cdot C_i, \qquad M = \frac{a^2}{v} - d. \tag{3}$$

All intersections in this display are taken on Y.

On X_t let g_t be the unique Ricci-flat Kähler metric with $[\omega_t] = H$. Let g_0 be the singular Ricci-flat Kähler metric in class H on $M_0 = X_0^{\mathrm{reg}}$, supplied by [EGZ09, HS17]. By central harmonic representatives we mean smooth L^2 two-forms on (M_0, g_0) that are closed and coclosed. Sections 3 and 6 construct the representatives used below and justify their domains and uniqueness. We use the nonnegative real Hodge Laplacian $\Delta_p = dd^* + d^*d$ on real p-forms and its ordinary L^2 norm. Write $\lambda_2^+(t)$ for its least strictly positive eigenvalue in degree two. The subscript indicates form degree. All limits below are along the fixed positive ray $0 < t < t_0$.

Theorem 1.1. *Under the preceding hypotheses, $M > 0$, and*

$$\lambda_2^+(t) = \frac{2\pi \sum_{i=1}^k b_i^2}{M\,|\log t|}(1 + o(1)). \tag{4}$$

For all sufficiently small t, this eigenvalue has real multiplicity one, and its real eigenspace is generated by a coexact primitive $(1,1)$-form. There is a constant $\gamma > 0$ such that the degree-one spectrum and every other positive degree-two eigenvalue are at least γ. Moreover,

$$\lambda_2^+(t) \leq \frac{2\pi \sum_i b_i^2}{M\,|\log t|} + C|\log t|^{-2}. \tag{5}$$

The constants refer to the fixed family, polarization, ray and local identifications. No uniformity as the leading coefficients in (1) degenerate is asserted.

Replacing D by $cD + eH$, with $c \neq 0$, multiplies both M and $\sum b_i^2$ by c^2. Thus the coefficient belongs to the one-dimensional divisor quotient modulo the surviving polarization. It is expressed in finite global intersection and restriction data; the list of local nodes alone does not determine it. The theorem gives a matching leading asymptotic and a sharper one-sided upper estimate. It does not assert a two-sided second-order remainder.

1.2. **The complete fixed-family cluster.** The same geometric problem allows several divisor directions to be lost. Retain the preceding family, polarization and simple-smoothing hypotheses, but replace (2) by $b_2(Y) = b$, $b_2(X_t) = s$ and $m = b - s > 0$. Set

$$V = H^2(Y;\mathbb{R}), \qquad r(D) = (D \cdot C_i)_i, \qquad K = \ker r, \qquad \mathcal{Q} = V/K.$$

The attachment sequence identifies K with the surviving cohomology $H^2(X_t;\mathbb{R})$. Define the forms

$$G(D,E) = -\int_Y DEH + \frac{3(\int_Y DH^2)(\int_Y EH^2)}{2\int_Y H^3}, \tag{6}$$
$$\mathcal{E}([D],[E]) = r(D)\cdot r(E).$$

Let $\mathcal{M}$ be the quotient form induced by G on $\mathcal{Q}$, equivalently the minimum squared norm over each coset. The theorem includes the positivity needed for this definition. Write $\lambda^+_{2,j}(t)$ for the ordered positive real degree-two eigenvalues, with multiplicity.

Theorem 1.2. *The form G is the positive-definite inner product of central harmonic representatives, and $\mathcal{E},\mathcal{M}$ are positive definite on the m-dimensional space $\mathcal{Q}$. If $0<\mu_1\le\cdots\le\mu_m$ are the generalized eigenvalues of $\mathcal{E}$ relative to $\mathcal{M}$, then*

$$\lambda^+_{2,j}(t) = \frac{2\pi\mu_j}{|\log t|}(1+o(1)), \qquad 1\le j\le m. \tag{7}$$

The degree-one spectrum and $\lambda^+_{2,m+1}(t)$ have a uniform positive lower bound. For small t, the entire m-dimensional positive small spectral space consists of coexact primitive real $(1,1)$-forms. A multiplicity-one μ_j gives an eventually simple eigenvalue. These assertions concern the fixed family and ray; no uniformity in degenerating node-scale ratios or two-sided next-order error is asserted.

The quotient removes all surviving harmonic directions, rather than only H. It is calculated by the intersection Schur complement (28). When $b=2$ and $s=1$, this gives the coefficient of Theorem 1.1; its one-sided sharper upper bound is proved separately. Section 6 supplies the full kernel projection and the ordered lower bounds needed for Theorem 1.2.

1.3. **Analytic inputs and proof.** Singular Ricci-flat metric existence, conical regularity and the local expansions for the actual smoothing metrics are supplied by [EGZ09, HS17, Ben26]. They do not determine the compact small-eigenvalue coefficients. We combine those metric results with conic spectral compactness and a sharp capacity calculation to prove (4) and the ordered matrix law (7).

Anné–Takahashi describe the spectral limits in exact-cone surgery in terms of the central operator, complete-cap harmonic forms and critical extended solutions [AT15]. We use their norm decomposition for finitely many caps, with the critical terms excluded in physical degrees one and two. The proof then has three steps. The restriction map fixes the harmonic link periods, and a scalar correction computes the central harmonic Gram form from intersections. On exactly conical models, logarithmic cutoffs give upper bounds and cap capacity gives matching lower bounds on every limiting spectral subspace. Dodziuk's comparison of Hodge eigenvalues under equivalent metrics transfers the entire ordered cluster to the actual Ricci-flat family [Dod82].

The leading law requires only the C^0 condition stated in Definition 4.2; its geometric verification is separate from the model argument. The differentiated estimate of Proposition 2.2 is used for the sharper rank-one upper remainder. Projection off all surviving harmonic classes is performed on the models, where the needed compact convergence follows from the fixed central elliptic problem.

1.4. **Three compact applications.** For independent linear forms l_1,l_2 and general quartics q_1,q_2 on $\mathbb{P}^4$, the split quintic $X_0=\{l_1q_2-l_2q_1=0\}$ has sixteen nodes on the plane $\{l_1=l_2=0\}$. Its incidence resolution lies in $\mathbb{P}^4\times\mathbb{P}^1$, with hyperplane classes H,L.

Corollary 1.3. *Let $X_t = \{l_1q_2 - l_2q_1 + tG = 0\}$, where the quintic G is nonzero at every node. For the Ricci-flat metrics in class H,*

$$\lambda_2^+(t) \sim \frac{10\pi}{|\log t|}.$$

The eigenvalue has all the simplicity, type, gap and one-sided remainder properties of Theorem 1.1.

Here $r(L) = (1, \dots, 1)$ and $M = 16/5$. The same primitive diagonal vector generates the integral relation among the cohomological vanishing cycles. The marked transition, including its full integral rings and monodromy image of index sixteen, is stated in Theorem A.1 and proved in Appendices A and B.

Eight nodes. Let $B = \mathbb{P}^1 \times \mathbb{P}^1$ and $Z = \mathbb{P}_B(\mathcal{O} \oplus \mathcal{O}(1,-1)) \times_B \mathbb{P}_B(\mathcal{O} \oplus \mathcal{O}(1,-1))$. In the quotient convention write f, g for the base hyperplane classes and ξ_1, ξ_2 for the tautological classes. The anticanonical contraction of a general anticanonical hypersurface $Y \subset Z$ is the eight-node threefold X_0 constructed in [W26e, Appendix A, Proposition A.1].

Corollary 1.4. *This transition admits polarized smoothings with trivial relative canonical bundle and first-order smoothing at every node, whose central polarization pulls back to $H = (4g + 2\xi_1 + 2\xi_2)|_Y$. For every such fixed family and ray, the actual Ricci-flat metrics in class H have precisely two small positive degree-two eigenvalues, with*

$$\lambda_{2,1}^+(t) \sim \frac{\pi}{|\log t|}, \qquad \lambda_{2,2}^+(t) \sim \frac{5\pi}{4|\log t|}. \tag{8}$$

Both are eventually simple, coexact and primitive real $(1,1)$. The degree-one spectrum and all other positive degree-two eigenvalues have a uniform positive lower bound for that fixed family.

Here $b_2(Y) = 4$ and $b_2(X_t) = 2$. The period matrix and the quotient mass are calculated from the earlier proposition's divisor intersections and two exceptional-curve types. Section 8 verifies the additional polarized-family hypotheses and proves the corollary. This is an application of Theorem 1.2; it makes no new integral saturation assertion for the eight-node family.

Four nodes. Let X_9' be the four-node partial resolution on the nonmirror side of [W26k, Section 11.1], with the chosen projective small resolution Y. Section 9 specifies the ample Cartier class $\mathcal{L}_0$ by the eleven-ray support vector (58). Write $H = c_1(q^*\mathcal{L}_0)$.

Corollary 1.5. *The threefold X_9' admits a relatively polarized smoothing carrying exactly $\mathcal{L}_0$, with trivial relative canonical bundle and first-order smoothing at every node. One has $b_2(Y) = 6$, $b_2(X_t) = 3$ and $H^3 = 8712$. For every such fixed family and ray, the actual Ricci-flat metrics in class H have precisely three small positive degree-two eigenvalues, with*

$$\begin{aligned} |\log t|\lambda_{2,1}^+(t) &\longrightarrow \frac{2\pi}{7}, \\ |\log t|\lambda_{2,2}^+(t) &\longrightarrow \frac{\pi(375 - 3\sqrt{3065})}{628}, \\ |\log t|\lambda_{2,3}^+(t) &\longrightarrow \frac{\pi(375 + 3\sqrt{3065})}{628}. \end{aligned} \tag{9}$$

All three are eventually simple, coexact and primitive real $(1,1)$*. The degree-one spectrum and all other positive degree-two eigenvalues have a uniform positive lower bound for that fixed family.*

The source's integral exceptional relation $(1,-1,-1,1)$ supplies a full-support deformation tangent through [W26j, Theorems 2.5 and 4.5]. We prove that the exact ample class extends, and calculate the full three-dimensional harmonic quotient. The resulting spectral law is an additional consequence on the established original-side geometry; it is not a claim about the mixed-singularity X_9 or the distinct mirror family.

1.5. **Relation to earlier work.** The split-quintic construction and conifold attachment sequences are classical [GMS95, Ros06, LLW18]. Our two toric applications use specific earlier geometric constructions: [W26e, Appendix A, Proposition A.1] supplies the eight-node contraction and Picard rank, while [W26k, Section 11.1] supplies the four-node threefold X_9' and its exceptional circuits. The nodal specialization of [W26j, Theorems 2.5 and 4.5] provides the latter's deformation tangent and smooth deformation base. In both applications we check first-order smoothing and extension of the specified ample class before applying the spectral theorem.

The curve-class map and the divisor-period map are transposes:

$$A : \mathbb{Q}^k \longrightarrow H_2(Y;\mathbb{Q}), \quad e_i \longmapsto [C_i], \qquad r = A^*, \qquad \operatorname{im} r = (\ker A)^{\perp}. \tag{10}$$

The space $\ker A$ governs localized deformation tangents, whereas $\ker r$ is the surviving second cohomology and $\operatorname{im} r$ supplies the spectral periods. The harmonic quotient also requires the polarized intersection form. Integral saturation is proved from the actual attachments where it is asserted; it does not follow from (10).

Anné–Takahashi's conic surgery theorem includes logarithmic modes [AT15]. Rowlett proves spectral and heat convergence under her asymptotically conic convergence hypotheses [Row08]. Guillarmou–Sher's low-energy resolvent theorem [GS15, Theorem 1] excludes the harmonic link two-form channel present here through its condition (4). For scalar Laplacians in different degenerations, Cao proves two-sided inverse-logarithmic orders for induced total-space metrics [Cao26, Theorems 0.2–0.4], and Dai–Yoshikawa obtain small-eigenvalue orders and the leading product in reduced curve degenerations [DY26, Theorems 0.2–0.3]. These results concern different operators and metrics. Here the conclusion is the individual leading coefficient of every lost-divisor mode for the actual polarized Ricci-flat metrics, computed from the full quotient harmonic norm.

The tangent connections in Appendix B are induced by those Ricci-flat metrics. Collins–Picard–Yau treat tangent-bundle HYM geometry on balanced conifold transitions [CPY24]; our appendix identifies the flat tangent-sheaf limit and its reflexive hull for the specified Kähler family.

Section 2 establishes the actual metric comparison. Section 3 calculates the harmonic norm and the integral restriction map. Sections 4–6 prove the model spectral law, transfer it using only C^0 comparison, and isolate the stronger rank-one remainder. Sections 7, 8 and 9 give the three applications. The appendices retain the supporting integral and tangent calculations.

2. Conifold geometry and actual metric comparison

2.1. **The link and its integral normalization.** An ordinary double point is analytically the cone $\sum_{j=1}^4 z_j^2 = 0$. Its Ricci-flat metric cone is $g_C = dr^2 + r^2 h$, where the homogeneous

metric on $T^{1,1} \simeq S^2 \times S^3$ is

$$h = \frac{1}{9}\eta_c^2 + \frac{1}{6}\sum_{j=1}^{2}(d\theta_j^2 + \sin^2\theta_j\, d\phi_j^2), \qquad \eta_c = d\psi + \cos\theta_1 d\phi_1 + \cos\theta_2 d\phi_2. \tag{11}$$

Here ψ has period 4π and $\mathrm{Ric}_h = 4h$. Put $\sigma_j = \sin\theta_j d\theta_j \wedge d\phi_j$ and

$$\vartheta = \frac{\sigma_1 - \sigma_2}{8\pi}. \tag{12}$$

The circle bundle over $\mathbb{P}^1 \times \mathbb{P}^1$ has Chern class $(1,1)$ up to a common sign. A primitive two-cycle projects with degrees $(1,-1)$; therefore ϑ has period one. It is harmonic: it is closed, its horizontal part is anti-self-dual, and wedging that part with $d\eta_c = -(\sigma_1 + \sigma_2)$ gives zero. In the orthonormal frame,

$$|\vartheta|_h^2 = \frac{9}{8\pi^2}, \qquad \mathrm{Vol}(T^{1,1}, h) = \frac{16\pi^3}{27}, \qquad c := \|\vartheta\|^2_{L^2(h)} = \frac{2\pi}{3}. \tag{13}$$

The selected small-resolution ruling fixes the sign of its primitive period. On a cone, ϑ is a primitive real $(1,1)$-form, of pointwise size $O(r^{-2})$.

The smoothing cap is T^*S^3 with the Stenzel metric, asymptotic with positive differentiated rate to this cone [Ste93, CO90]. The small-resolution cap is the total space of $\mathcal{O}_{\mathbb{P}^1}(-1)^{\oplus 2}$ with its Ricci-flat resolved-conifold metric. The two caps have the same link but retain different primitive cohomology: restriction to the link is an isomorphism in degree three for the smoothing cap and in degree two for the resolution cap.

2.2. **The central metric and the smoothing scale.** Let X_0^{reg} carry the singular Ricci-flat Kähler form ω in class H. The bounded-potential construction [EGZ09, §7.2, Theorem 7.5 and Corollary 7.6] gives a closed positive pullback current on Y. Its Bedford–Taylor products have their cohomological masses and give no mass to the exceptional analytic set. The conical theorem of Hein–Sun [HS17, Theorem 1.4 and Corollary 1.5] supplies holomorphic node charts such that, for some $\mu > 0$,

$$|\nabla_C^j(g_0 - g_C)|_{g_C} \le C_j r^{\mu - j} \quad (j \ge 0). \tag{14}$$

Use the minimum rate over the finitely many ends. Benabida's Theorem A provides the corresponding polyhomogeneous expansion [Ben26]. The cited metric theorems use the class $2\pi c_1(\mathcal{L})$ for the polarizing line bundle $\mathcal{L}$. We rescale their metrics by $(2\pi)^{-1}$ to obtain our class $H = c_1(\mathcal{L})$, with the corresponding fixed rescaling of cone radii and cap normalizations. Written as $dr^2 + r^2 h$, the normalized link metric and (13) are unchanged.

Condition (1) makes the total-space projection a holomorphic Morse function near each node. The local family therefore has the weighted ordinary-node model, with weights $3/2$ on the coordinates and 3 on the algebraic smoothing parameter. This model is compatible with the central conical chart. Indeed, the transition between two ordinary-quadric charts lifts to an ambient holomorphic germ Φ with $Q \circ \Phi = uQ$, where $Q = \sum z_j^2$ and u is a holomorphic unit. Its linear part A satisfies $Q(Az) = u(0)Q(z)$ and extends to the weighted smoothing after a fixed rescaling of the base. Since all coordinate weights are $3/2$, the terms $\Phi - A = O(|z|^2)$ have relative conic order $O(r^{3/2})$, with differentiated bounds. After this linear normalization, (14) therefore holds in the family chart, with μ decreased if necessary. Along the fixed positive ray we use the reference metric scale

$$\epsilon = t^{1/3}. \tag{15}$$

On the smoothing branch, metrics, Kähler forms and eigenvalues indexed by ϵ are evaluated at the algebraic parameter $t = \epsilon^3$. The individual cap scales have fixed positive leading ratios to ϵ. Multiplying a local normalization by a fixed constant changes a logarithmic length by a bounded term.

Lemma 2.1 (Uniform corner extraction)**.** *Let $f(\rho_1, \rho_2, y)$ be a polyhomogeneous tensor coefficient on a compact product corner. Suppose its index sets at both faces have only log-free order-zero terms and terms of strictly positive real order, and let $f_0(\rho_1, y)$ be its trace at $\rho_2 = 0$. There is $a_0 > 0$ such that, for every fixed number of b-derivatives,*

$$|D_b^j(f - f_0)| \le C_j \rho_2^{a_0} \tag{16}$$

uniformly up to $\rho_1 = 0$.

Proof. Discreteness of the index set gives a positive gap above zero at the ρ_2 face. Choose a_0 below that gap and a positive remainder order. Expansion coefficients and their fixed b-derivatives are bounded at the other face, since its index set has no negative orders or order-zero logarithms. Subtracting the trace removes the order-zero term; positive powers absorb the remaining finite logarithmic powers. The conormal remainder has the same bound. A finite corner cover makes the constants uniform. Pointwise convergence at fixed $\rho_1 > 0$ alone would not imply this conclusion. □

Proposition 2.2 (Actual metric comparison)**.** *There are collar-compatible exterior identifications F_ϵ and constants $a_0 > 0$, $r_0 > 0$ such that, for fixed sufficiently large A,*

$$|\nabla_C^j (F_\epsilon^* g_\epsilon - g_0)|_{g_C} \le C_j (\epsilon/r)^{a_0} r^{-j}, \quad A\epsilon \le r \le r_0, \quad j = 0, 1. \tag{17}$$

The same comparison holds for the Kähler forms. On fixed exterior compact sets the C^1 differences are $O(\epsilon^{a_0})$. On fixed compact subsets of each rescaled cap, the metrics converge in all derivatives to the normalized Stenzel metric.

Proof. We use the construction of [Ben26, Theorem C], with the following finite-node extension and tensor deduction. The cited work is published; our numbered references to its proof refer to arXiv version 3, as specified in the bibliography. Remark 5.2 there describes the extension, but its Lemma 5.4 is formulated with one cap face. We give the global estimate needed to replace that hypothesis.

The normalized scalar inverse. Let ρ_ϵ be a positive radius, equal to the central radius on the necks and comparable to ϵ on the cores. For $-4 < \beta < 0$ put

$$\|u\|_{H^q_\beta}^2 = \sum_{j=0}^{q} \int |\rho_\epsilon^{j-\beta} \nabla^j u|^2 \rho_\epsilon^{-6} \, dV.$$

The CS space in [Pac13, Definitions 11.1 and 11.6] is the connected central regular locus; the AC space is the disjoint union of the normalized Stenzel caps. Disconnected AC spaces are allowed. All ends are filled at scale ϵ, with fixed ratios absorbed into the caps. Choose a smooth density ζ of integral one supported in a fixed central compact set. Theorem 12.3(1) of [Pac13] gives the estimate

$$\|u\|_{H^{q+2}_\beta} \le C_q \|\Delta_\epsilon u\|_{H^q_{\beta-2}}, \qquad u \in E_\epsilon := \left\{ \int \zeta u = 0 \right\}, \tag{18}$$

uniformly in ϵ, for its parametric connected-sum metric. The complement is precisely the pullback of the fixed central complement in that proof: the cutoff used there is one on $\operatorname{supp} \zeta$. The restricted Laplacian has the same image as the full Laplacian. On the connected closed

smoothing this image is $\{f : \int f\, dV_\epsilon = 0\}$, and the kernel is the single space of constants. Thus the range condition is global; there is no separate normalization or mean-zero condition for each cap.

The initial Kähler tensor in [Ben26, Proposition 5.9] has the prescribed central and cap traces with positive orders at the opposite faces. Compare it with Pacini's metric, interpolated at $r = \epsilon^\tau$, $0 < \tau < 1$. On the central and cap sides, its initial face expansions give errors bounded by positive powers of $\epsilon^{1-\tau}$ and ϵ^τ. On the intervening annuli, (14) and the AC rate give the same conclusion in every fixed scaled differentiability norm. Therefore

$$\|\Delta_{\text{initial}} - \Delta_{\text{Pacini}}\|_{H^{q+2}_\beta \to H^q_{\beta-2}} = o(1).$$

Absorption in (18) proves the estimate for the initial family and its subsequent boundary-vanishing formal corrections. This comparison uses the initial compactification, not any estimate for the final Ricci-flat metric. Theorem 11.7 of [Pac13] supplies the uniform scaled Sobolev and multiplication estimates used in the nonlinear argument.

Finite-node formal corrections. The surgery space has one connected central face and finitely many pairwise disjoint cap faces. The central coefficient problems of [Ben26, §§5–6] are scalar equations on that one connected conifold, with one integral compatibility condition. The finite-end positive-weight inverse includes the low harmonic terms at every end [Ben26, Proposition 3.4]. These terms cannot be discarded when differentiating the potentials. For the node, the first scalar link eigenvalues are $0, 33/4, 12$, with growth rates $0, 3/2, 2$; below growth two the terms are constants and the real and imaginary parts of the four holomorphic linear coordinates [HS17, Theorem 2.14(1), Example 2.16]. Extend the corresponding central holomorphic germs through the smoothing charts and multiply by disjoint cutoffs equal to one near the nodes. Their fibrewise $\partial\bar\partial$ vanishes near the corners, and all cutoff derivatives have fixed exterior support. Consequently the low terms introduce no negative corner tensor order.

The AC coefficient equations are a finite direct sum of the inverses in [Ben26, Proposition 3.6]. Their decaying weight range has no componentwise integral obstruction. The global volume normalization gives the compatibility for the central equation. More explicitly, each cap's volume starts at order ϵ^6, so a central residual coefficient of order less than six has zero integral by the push-forward argument of [Ben26, Proposition 5.5]. At the next stage one improves the central residual to almost order six and the cap residuals to almost order four, factors out $\epsilon^{4-\eta}$ for a sufficiently small $\eta > 0$, and repeats the same procedure. Multiplication by this scalar preserves the global zero-integral identity. Cap contributions add in that identity; they are not required to vanish individually. Finitely many disjoint faces permit common choices of weights and the same Borel summation.

At a product corner write $\epsilon = \rho_1\rho_2$. The tensor orders in [Ben26, proof of Proposition 6.1] are the following. A central potential correction $\epsilon^\delta w$, after removing the pluriharmonic terms just treated, has w of weight $a > 2 - \delta$. Choose $\max(\delta/2, 2 - a) < \delta' < \delta$. Its twice-differentiated conic tensor has order

$$\rho_2^{\delta'}\rho_1^{a-2+\delta'}.$$

Both exponents are positive, logarithms are absorbed by the loss in δ', and $2\delta' > \delta$ improves the nonlinear residual. A cap correction has tensor order $\rho_1^{\delta'}\rho_2^{a+2+\delta'}$, with positive exponents; the cap weight choice preserves the previously improved central order. These are tensor estimates, not merely positive orders for potentials.

The inverse (18) and the scaled nonlinear bounds give the final correction as in [Ben26, §7]. With the same normalization for every order, uniqueness and Sobolev embedding give

$$|\nabla^j \phi_\epsilon| \le C_{N,j} \epsilon^N \rho_\epsilon^{\beta - j} \quad \text{for every fixed } N, j.$$

Since $\epsilon = \rho_1 \rho_2$ and ρ_ϵ is comparable to ρ_1, its metric correction is flat at both faces after any fixed number of conic derivatives. Joint flatness follows by differentiating the normalized parameter equation: the coefficient derivatives cost only fixed powers, and the same uniform linearized inverse gives the displayed arbitrary-order decay inductively. Thus fibrewise estimates are not being used alone to infer transverse regularity.

The final tensors consequently have nonnegative polyhomogeneous index families in the vertical conic bundles: the order-zero term at each face is log-free, and every other exponent has positive real part. The face values asserted in Theorem C are unchanged. The central value is the full g_0; on a cap, $\epsilon^{-2} g_\epsilon$ restricts to its normalized Stenzel metric. In the conic bundle the latter is $\rho_2^2 g_{\mathrm{St}}$ in the cap b-frame. Since $\rho_2 = (1 + \widetilde{r}^2)^{-1/2}$, it is bounded up to the corner and has the cone trace there.

Extraction in the conic frame. The index families just obtained satisfy Lemma 2.1. Map $(y, r, 0)$ to $(y, r, \epsilon/r)$, taking $r = \rho_1$. The vertical b-frame is $\rho_1 \partial_{\rho_1} - \rho_2 \partial_{\rho_2}, \partial_y$. After division by ρ_1 its pullback is the cone frame $\partial_r, r^{-1} \partial_y$. A cone covariant derivative costs r^{-1} and bounded b-differentiation. Equation (16) gives (17). Extend the prescribed collar lifts by a partition of unity on the smooth fixed exterior; their flow gives compatible identifications. There is no holomorphicity requirement. The exterior and rescaled-cap estimates follow from the respective face restrictions. The same argument applies to the Kähler tensors.

Finally, the initial form in [Ben26, Proposition 5.9] is obtained from curvature on the actual polarizing line bundle. The Stenzel form has a global potential, and the gluing and all subsequent corrections are fibrewise $\partial\bar{\partial}$-exact. The construction therefore stays in $2\pi c_1(\mathcal{L}_t)$. Ricci-flat uniqueness in this class, followed by the fixed rescaling already specified, identifies it with g_t on the prescribed complex family. □

The trace in (16) is the full central tensor. Comparing both metrics separately with the cone would leave an $O(r^\mu)$ term independent of ϵ; that weaker estimate is not used for the differentiated remainder argument.

3. Harmonic representatives and intersection mass

3.1. **Central cohomology and harmonic representatives.** Write $M_0 = X_0^{\mathrm{reg}} = Y \setminus \bigcup_i C_i$. Removing a submanifold of real codimension four does not change cohomology in degrees one and two. The Thom sequence consequently gives

$$H^1(M_0; \mathbb{R}) = 0, \qquad H^2(M_0; \mathbb{R}) = H^2(Y; \mathbb{R}) = V. \tag{19}$$

The common exterior has boundary a disjoint union of copies of $S^2 \times S^3$. In the smoothing each filling has zero second cohomology. Mayer–Vietoris therefore identifies the image of $H^2(X_t)$ in $H^2(M_0)$ with the kernel of restriction to these links. For the general hypotheses of Theorem 1.2, this image is $K = \ker r$ and has dimension s. In the rank-one setting it is $\mathbb{R}H$; then a class D independent of H has nonzero restriction vector (b_i). Indeed every b_i is nonzero in that setting: otherwise the effective curve C_i would pair trivially with all divisor classes on projective Y.

Van Kampen gives simple connectivity of X_t, since the links and both fillings are simply connected. For either compact Kähler threefold, $b_1 = 0$ implies $H^1(\mathcal{O}) = 0$. Serre duality

and $K \simeq \mathcal{O}$ give $H^2(\mathcal{O}) = 0$, and Hodge symmetry gives $H^{2,0} = 0$. The integral second-cohomology classes are consequently divisor classes. For the next proposition take any integral D independent of H, with a, d, v, b_i, M as in (3).

Proposition 3.1. *There is a real closed $(1,1)$-form η on M_0 representing D, with $\Lambda_\omega \eta = 3a/v$, such that*

$$\alpha = \eta - \frac{a}{v}\omega \tag{20}$$

is closed, coclosed, primitive and in the Friedrichs form domain. For some $\nu > 0$ its expansion on end i is

$$\alpha = b_i \vartheta_i + O(r^{\nu-2}), \qquad \nabla_C^j(\alpha - b_i\vartheta_i) = O(r^{\nu-2-j}), \tag{21}$$

for every $j \geq 0$. Its norm and orthogonality are

$$\|\alpha\|^2_{L^2(g_0)} = \frac{a^2}{v} - d = M > 0, \qquad \langle \alpha, \omega \rangle_{L^2} = 0. \tag{22}$$

Proof. We use $d^c = \frac{i}{2}(\bar\partial - \partial)$, so $dd^c = i\partial\bar\partial$. Choose a smooth Hermitian metric on the line bundle with class D on Y, and let θ_b be its real normalized curvature form. A local small-resolution neighborhood retracts onto its exceptional $\mathbb{P}^1$. The node is rational, and the higher direct images of its structure sheaf vanish. On the inverse image of a sufficiently small Stein neighborhood, $H^1(\mathcal{O}) = H^2(\mathcal{O}) = 0$; equivalently one can use the infinitesimal-neighborhood filtration with quotients $\mathcal{O}(n)^{\oplus(n+1)}$. The exponential sequence identifies the local Picard group with $\mathbb{Z}$, by exceptional degree. Thus the line bundle is the b_ith power of the selected ruling bundle.

The normalized curvature $\sigma_1/(4\pi)$ of that ruling differs from (12) by $(\sigma_1+\sigma_2)/(8\pi)$, a constant multiple of $dd^c \log r$. Changing the local Hermitian metric therefore makes its curvature $b_i\vartheta_i$. The local logarithmic metric ratios u_i satisfy $u_i = O(1 + |\log r|)$ and $\nabla_C^j u_i = O(r^{-j})$ for $j \geq 1$. Patch them with disjoint cutoffs. Then

$$\theta = \theta_b + dd^c u$$

is a global real closed $(1,1)$-form representing D, equal to $b_i\vartheta_i$ near each end. The original smooth curvature has cone size $O(r^{-2})$ with differentiated bounds. Since the cone charts are holomorphic, type is preserved in this construction. Cone primitivity and (14) give $\Lambda_\omega \theta = O(r^{\mu-2})$.

The bounded-potential intersection products on Y give

$$\int_{M_0} \omega^3 = v, \qquad \int_{M_0} \theta_b \omega^2 = a, \qquad \int_{M_0} \theta_b^2 \omega = d. \tag{23}$$

For the mixed products, pairing the closed current with smooth closed forms gives their cohomological values by Stokes. The products have no mass on the exceptional curves. Positivity of θ_b is unnecessary: write it as the difference of two smooth positive forms, using a large multiple of a fixed Kähler form, and apply linearity to the bounded- potential products [BT82, EGZ09].

Truncate every end at $r = \rho$. The transgression for replacing θ_b by θ in the second integral of (23) is $d^c u \wedge \omega^2$, whose boundary integral is $O(\rho^4)$. In the third integral it is $d^c u \wedge (\theta + \theta_b) \wedge \omega$, with boundary integral $O(\rho^2)$. Both vanish. The bulk terms are absolutely integrable, since their worst size is $O(r^{-4})$ against volume $O(r^5 dr)$. Consequently $\int \theta\omega^2 = a$ and $\int \theta^2\omega = d$.

Choose $0 < \nu < \min(\mu, 1)$ away from scalar indicial roots. The conical scalar Poisson isomorphism [Ben26, Proposition 3.4], with one cutoff constant for each end, solves

$$\Lambda_\omega dd^c f = \frac{3a}{v} - \Lambda_\omega \theta, \qquad \int_{M_0} f\, dV = 0. \tag{24}$$

Here the right side has zero integral because $\Lambda\theta\, dV = \theta\omega^2/2$ and $\mathrm{Vol}(M_0) = v/6$. It has weighted order $O(r^{\nu-2})$. The operator on the left is $-\Delta_0/2$ in our convention. The link Ricci bound implies that the first positive scalar homogeneous rate is at least one; below ν only constants occur. The solution therefore has $f = f_i + O(r^\nu)$, with differentiated bounds on each end. The finite-end version is the direct sum of these constant asymptotic terms.

Put $\eta = \theta + dd^c f$. It has the required constant contraction, and (20) is primitive and closed. The Kähler identity $*\alpha = -\alpha \wedge \omega$ implies coclosedness. The new transgressions are $d^c f \wedge \omega^2$ and $d^c f \wedge (\eta + \theta) \wedge \omega$, with boundary integrals $O(\rho^{\nu+4})$ and $O(\rho^{\nu+2})$. Hence

$$\int \eta\omega^2 = a, \qquad \int \eta^2\omega = d,$$

and the primitive star identity proves

$$\|\alpha\|^2 = -\int \Big(\eta - \frac{a}{v}\omega\Big)^2 \omega = -d + \frac{a^2}{v}.$$

Primitivity gives pointwise orthogonality to ω. The nonzero class $D - (a/v)H$ implies $\alpha \neq 0$, so this norm is positive. The expansion (21) follows from the scalar estimate; the bounded term proportional to ω is absorbed by $\nu < 1$.

Finally, let χ_ρ vanish for $r \leq \rho^2$, equal one for $r \geq \rho$, and interpolate smoothly with $|\chi_\rho'| \leq C/(r|\log\rho|)$. The removed L^2 mass is $O(\rho^2)$. Since $d\alpha = d^*\alpha = 0$,

$$\|d(\chi_\rho\alpha)\|^2 + \|d^*(\chi_\rho\alpha)\|^2 \leq \frac{C}{|\log\rho|^2}\int_{\rho^2}^{\rho} \frac{dr}{r} \longrightarrow 0.$$

This approximates α in the closed Friedrichs quadratic-form norm by compactly supported forms and proves the domain assertion. In particular α is in the Friedrichs kernel. The same argument applies to ω. □

In the rank-one setting, (22) is the quotient norm after removing the surviving Kähler direction. This identifies the global mass needed in the spectral comparison without leaving an unspecified analytic coefficient among the input data.

3.2. **The full harmonic space and its quotient norm.** Put $V = H^2(Y;\mathbb{R})$ and let $r : V \to \mathbb{R}^k$ be the exceptional-period map. The Thom and Mayer–Vietoris arguments of Proposition 3.3 give

$$K := \ker r = \mathrm{im}\big(H^2(X_t;\mathbb{R}) \longrightarrow V\big), \qquad \dim K = s.$$

In particular $H \in K$. Write

$$B(D,E) = \int_Y DEH, \qquad a(D) = \int_Y DH^2, \qquad v = \int_Y H^3. \tag{25}$$

Simple connectivity transfers to the smoothing by van Kampen, as before. Both smooth spaces have $H^{2,0} = 0$, so integral second-cohomology classes are represented by line bundles.

Apply the curvature construction and scalar correction in the proof of Proposition 3.1 to an integral basis of V, and extend linearly. For $D \in V$ this gives a closed and coclosed real $(1,1)$-form η_D with class D, constant contraction $3a(D)/v$, and link coefficients $r_i(D)$. The scalar inverse is fixed and linear; its solvability condition uses $a(D)$. The local Picard

calculation, transgression estimates and logarithmic domain cutoffs do not use the dimension of V. After taking the minimum of finitely many positive rates, the resulting estimates hold uniformly on bounded subsets of V. Polarizing (22) gives

$$G(D,E) := \langle \eta_D, \eta_E \rangle = -B(D,E) + \frac{3a(D)a(E)}{2v}. \tag{26}$$

Indeed the primitive parts have pairing $-B(D,E) + a(D)a(E)/v$, and $\|\omega\|^2 = v/2$. The independent classes of the η_D make G positive definite.

We record uniqueness for closed and coclosed L^2 representatives.

Lemma 3.2. *Let a smooth metric on M_0 be uniformly equivalent near each end to the six-dimensional cone metric. Every smooth globally exact L^2 two-form u has an L^2 primitive in* $\operatorname{Dom} d_{\min}$. *Consequently a smooth L^2 closed and formally coclosed two-form is determined by its ordinary de Rham class.*

Proof. On one end write $u = a(r) + dr \wedge b(r)$ in fixed link coordinates. Choose v_0 with $d_L v_0 = a(r_0)$ and set

$$\beta(r) = v_0 + \int_{r_0}^{r} b(x)\, dx.$$

Closedness gives $d\beta = u$. In the fixed link norm put $B_0(x) = \|b(x)\|_{L^2(L,h)}$. Cone equivalence then implies $\int_0^{r_0} x^3 B_0(x)^2 dx < \infty$. For $0 < r < \delta < r_0$, Cauchy–Schwarz gives

$$r\|\beta(r)\|_{L^2(L)} \le rC_\delta + 2^{-1/2} \left(\int_0^{\delta} x^3 B_0(x)^2 dx \right)^{1/2}.$$

First let $r \to 0$, then $\delta \to 0$. Thus $r\|\beta(r)\|_{L^2(L)} \to 0$. The one-form β is L^2, since its squared cone norm has radial weight r^3. If χ_ρ vanishes below ρ and equals one above 2ρ, with derivative bounded by C/ρ, then

$$\|d\chi_\rho \wedge \beta\|^2 \le C\rho^{-2} \int_{\rho}^{2\rho} r^3 \|\beta(r)\|^2_{L^2(L)} dr = o(1).$$

The L^2 tails of β and $d\beta = u$ also vanish.

Let w_0 be a global smooth primitive. On the end $w_0 - \beta$ is closed and hence exact, since $H^1(L) = 0$. Subtracting the differential of a cutoff potential patches w_0 to β near each tip without changing its differential. The resulting global w is L^2, and the preceding cutoff estimate proves $w \in \operatorname{Dom} d_{\min}$. If u is formally coclosed, testing on the compactly supported core and passing to its graph closure gives $u \in \operatorname{Dom} d^*_{\min}$ and $d^*_{\min} u = 0$. Hence $\|u\|^2 = \langle d_{\min} w, u \rangle = 0$. Apply this to the difference of two forms to obtain uniqueness. □

For the actual central metric, this lemma and the b constructed forms identify the smooth L^2 harmonic two-form space with V. Give $\mathcal{Q} = V/K$ the quotient norm $\mathcal{M}$ induced by G. The G-orthogonal lift $D^\perp$ of $[D]$ is primitive, because $G(D,H) = a(D)/2$. Thus

$$\mathcal{M}([D],[E]) = -B(D^\perp, E^\perp), \qquad D^\perp \perp_B K. \tag{27}$$

The restriction of B to K is nondegenerate: its decomposition $K = \mathbb{R}H \oplus (K \cap \ker a)$ has one positive direction and a negative-definite primitive part. In a basis of K followed by quotient lifts, the mass matrix is therefore

$$\mathsf{M} = -B_{DD} + B_{DK} B_{KK}^{-1} B_{KD}. \tag{28}$$

The period form $\mathcal{E}([D],[E]) = r(D) \cdot r(E)$ is positive definite on $\mathcal{Q}$. Subtracting the polarization component alone would not give this quotient mass when $s > 1$.

3.3. **The restriction map in topology and in the spectral law.** The attachment sequence below is classical conifold topology [Ros06, LLW18]; compare the integral exterior sequence in [W26k, Section 9.7]. We retain its marked integral maps to identify the same divisor periods in the spectral coefficient. The curve-class map used in the smoothing criteria of [W26i, W26j] is its dual, as in (10). No mirror identification is used here. Let U be a compact truncation of M_0, with marked boundary $\coprod_i \mathcal{L}_i$. Let F_i be the smoothing filling, a disk cotangent bundle of S^3, and write δ_i for its oriented zero section in X_t. We use integral cohomology until real coefficients are specified. A left Poincaré dual is characterized by $\int_{X_t} \mathrm{PD}_L(\delta_i) \wedge \zeta = \int_{\delta_i} \zeta$. The integral class itself is defined by the oriented Thom class, so this convention also applies when its global image is torsion.

Proposition 3.3 (Restriction and spectral coefficient)**.** *Under the geometric hypotheses of Theorem 1.2, the actual attachments give the integral exact sequence*

$$0 \longrightarrow H^2(X_t;\mathbb{Z}) \longrightarrow H^2(Y;\mathbb{Z}) \xrightarrow{r} \mathbb{Z}^k \xrightarrow{\partial} H^3(X_t;\mathbb{Z}) \longrightarrow H^3(U;\mathbb{Z}) \longrightarrow 0, \tag{29}$$
$$r(D) = (D \cdot C_i)_i, \qquad \partial e_i = \mathrm{PD}_L(\delta_i).$$

Thus $\operatorname{im} r$ *is exactly the group of integral relations between the cohomological vanishing classes. In the rank-one case (2),* $\ker(r \otimes \mathbb{R}) = \mathbb{R}H$*. On its one-dimensional quotient* $\mathcal{Q} = H^2(Y;\mathbb{R})/\mathbb{R}H$*, put*

$$P_H D = D - \frac{DH^2}{H^3} H, \qquad m_H([D]) = -\int_Y (P_H D)^2 H.$$

The form m_H *is positive definite, and the induced restriction* $r : \mathcal{Q} \to \mathbb{R}^k$ *satisfies*

$$\lim_{t \downarrow 0} |\log t| \lambda_2^+(t) = \frac{2\pi \|r([D])\|^2}{m_H([D])} \quad \text{for every } [D] \neq 0. \tag{30}$$

Here the Euclidean norm uses the primitive integral link generators. The same restriction coordinates are the boundary coefficients of the central primitive harmonic form used in the spectral proof.

Proof. The integral Thom isomorphism gives $H^j(Y,U;\mathbb{Z}) = \bigoplus_i H^{j-4}(C_i;\mathbb{Z})$. Consequently $H^2(Y;\mathbb{Z}) \to H^2(U;\mathbb{Z})$ is an isomorphism. The resolution filling retracts onto C_i, and its restriction in degree two to $\mathcal{L}_i$ is an isomorphism. Thus its marked boundary coefficient is $D \cdot C_i$.

Apply Mayer–Vietoris to $X_t = U \cup \coprod_i F_i$, with difference $j_U^* - j_F^*$. Since $H^1(\mathcal{L}_i) = H^2(F_i) = 0$, its degree-two part has the initial terms in (29). In degree three the filling restriction is an integral isomorphism, so the next difference map is

$$H^3(U;\mathbb{Z}) \oplus \mathbb{Z}^k \longrightarrow \mathbb{Z}^k, \qquad (y,z) \longmapsto r_3(y) - z.$$

Its kernel is the graph of r_3, canonically isomorphic to $H^3(U;\mathbb{Z})$ by projection. This proves exactness through the final exterior restriction. For the connecting class at the ith link, naturality factors the map through

$$H^2(\mathcal{L}_i;\mathbb{Z}) \longrightarrow H^3(F_i,\mathcal{L}_i;\mathbb{Z}) \longrightarrow H^3(X_t;\mathbb{Z}).$$

The first arrow takes the marked fibre-sphere generator to the integral Thom class of the zero section. The second is extension by excision. Orienting δ_i by the chosen collar convention identifies its image with $\mathrm{PD}_L(\delta_i)$. This proves the relation assertion integrally, without discarding torsion.

For the remaining assertions impose (2). After passage to real coefficients, the image of $H^2(X_t)$ is one-dimensional and contains H, hence is $\mathbb{R}H$. The projection $P_H D$ is unchanged by adding a multiple of H. Proposition 3.1 represents it by α and proves

$$m_H([D]) = \|\alpha\|^2 = \frac{(DH^2)^2}{H^3} - D^2H > 0.$$

On every link section, $[\alpha|_{\mathcal{L}_i}] = (D \cdot C_i)[\vartheta_i]$. Since $*_h\vartheta_i$ is closed, an exact discrepancy does not contribute to its pairing. Therefore, with $c = 2\pi/3$,

$$c^{-1} \int_{\mathcal{L}_i} \alpha|_{\mathcal{L}_i} \wedge *_h\vartheta_i = D \cdot C_i. \tag{31}$$

This is exactly the scalar boundary projection in Lemma 4.1. The upper and lower bounds proving Theorem 1.1 use these coefficients, normalized by the same harmonic mass. Substitution gives (30). Scaling $[D]$ multiplies its numerator and denominator by the same square. □

The proposition identifies the topological data in the spectral theorem; the analytic conclusion still requires its full metric and compactness arguments. In the general integral sequence, neither $\mathbb{Z}^k/\operatorname{im} r$ nor $H^3(U;\mathbb{Z})$ has been assumed torsion-free. Primitivity of the vanishing lattice is a separate conclusion in the split-quintic calculation below.

4. Logarithmic capacity and conic models

4.1. **A sharp cap estimate.** Let (Z, g) be an oriented complete six-manifold with one asymptotically conical end, with cone $dr^2 + r^2h$ and

$$|\nabla_C^j(g - g_C)|_{g_C} \le C_j r^{-a_1-j}, \qquad j = 0, 1, \qquad a_1 > 0.$$

Let ϑ be a harmonic two-form on the link and write $c = \|\vartheta\|^2_{L^2(h)} > 0$. Assume that the class of $*_h\vartheta$ extends to a closed three-form on a compact truncation Z_A. For a smooth two-form u on Z_T, let u_T denote its tangential boundary trace and set

$$B_u(T) = c^{-1} \int_L u_T \wedge *_h\vartheta, \qquad \operatorname{Cap}(T) = \inf_{B_u(T)=1} \int_{Z_T} (|du|^2 + |d_g^* u|^2)\, dV_g.$$

The infimum is over all smooth competitors; no elliptic boundary condition or existence of a minimizer is assumed.

Lemma 4.1. *For this fixed cap,*

$$\operatorname{Cap}(T) = \frac{c}{\log T} + O((\log T)^{-2}). \tag{32}$$

More precisely, every competitor satisfies

$$\int_{Z_T} (|du|^2 + |d_g^* u|^2)\, dV_g \ge \frac{c\,|B_u(T)|^2}{\log T + C_A} \tag{33}$$

for a fixed C_A and sufficiently large T.

Proof. Choose a closed three-form Ψ on Z_A whose boundary value is exactly $*_h\vartheta$. To obtain the exact value from a cohomological extension, subtract the discrepancy by an exact collar correction. Stokes and Cauchy–Schwarz give

$$cB_u(A) = \int_{Z_A} du \wedge \Psi, \qquad E_A(u) \ge \kappa_A |B_u(A)|^2, \quad \kappa_A = \frac{c^2}{\|\Psi\|^2_{L^2(Z_A)}} > 0.$$

On the end write $u = u_T(r) + dr \wedge v(r)$ and $B(r) = B_u(r)$. The projection of the radial component of du onto $dr \wedge \vartheta$ is $B'(r)dr \wedge \vartheta$, because $d_L^*\vartheta = 0$. Hence

$$\int_{A<r<T} |du|^2_{g_C} dV_C \geq c \int_A^T r|B'(r)|^2 dr.$$

The factor r follows from $|dr \wedge \vartheta|^2 = r^{-4}|\vartheta|^2_h$ and the six-dimensional volume density. For large fixed A, the AC comparison gives $w(r) = 1 - Cr^{-a_1} \geq 1/2$ such that

$$E_T(u) \geq \kappa_A B(A)^2 + c \int_A^T rw(r)|B'(r)|^2 dr.$$

Applying Cauchy–Schwarz to $B(T) = B(A) + \int_A^T B'(r)dr$ gives

$$|B(T)|^2 \leq \left(\kappa_A^{-1} + c^{-1} \int_A^T \frac{dr}{rw(r)}\right) E_T(u).$$

The radial integral is $\log(T/A) + O_A(1)$, which proves (33).

For the upper estimate, choose a smooth f vanishing near A and agreeing with $\log(r/A)$ for $r \geq 2A$. Extend $u = f(r)\vartheta/\log(T/A)$ by zero across the core. Its boundary coefficient is one and its exterior-differential energy is $c/\log(T/A) + O_A(\log^{-2} T)$. The metric error has integrable radial weight r^{-1-a_1} after the logarithmic factor is removed. For the codifferential, the cone identities $d_C^*\vartheta = 0$ and $\iota_{\nabla_C f}\vartheta = 0$ imply

$$|d_g^*\vartheta| \leq Cr^{-3-a_1}, \qquad |\iota_{\nabla_g f}\vartheta| \leq C|f'|r^{-2-a_1}.$$

The product rule consequently gives

$$\|d_g^* u\|^2 \leq \frac{C_A}{\log^2(T/A)} \left(1 + \int_{2A}^T (1 + \log(r/A))^2 r^{-1-2a_1} dr\right) = O_A(\log^{-2} T).$$

This proves the matching upper estimate. □

For the Stenzel cap the extension condition holds: a compact truncation is the disk cotangent bundle of S^3, and restriction $H^3(Z_A; \mathbb{R}) \to H^3(S^2 \times S^3; \mathbb{R})$ is an isomorphism. With the primitive normalization (13), $c = 2\pi/3$. Under $g \mapsto \epsilon^2 g$ the energy of two-forms is unchanged. Indeed the du term scales by $\epsilon^{6-2\cdot 3} = 1$; the other term has the same factor since $d^*_{\epsilon^2 g} = \epsilon^{-2} d_g^*$. An outer physical radius R therefore gives capacity

$$\frac{2\pi}{3\log(R/\epsilon)} + O(\log^{-2}(R/\epsilon)). \tag{34}$$

4.2. **Fixed conic models.** On the cap end choose the radial variable $1/\rho_2$ from the product corner in Section 2. In the notation of [Ben26, §5.3.2], its restriction is $\sqrt{1 + \widetilde{r}^2}$, where $\widetilde{r}$ is the original cap radius. This changes that radius by a relative $O(\widetilde{r}^{-2})$ term, with differentiated bounds, and preserves the asymptotically conical estimates after decreasing their positive decay rate if needed. Fix a small central radius $R > 0$ and a large cap radius $A > 0$. Let g_0^R equal g_0 outside $r = 2R$, equal g_C inside $r = R$, and interpolate between them. Let g_{St}^A agree with the normalized Stenzel metric up to radius A, with g_C beyond $2A$, and interpolate on the intervening annulus. These are smooth positive metrics; they need not be Kähler. Their relative errors are respectively $O(R^\mu)$ and $O(A^{-a_1})$.

For $2A\epsilon < R$, glue the fixed central metric to the scaled metrics $\epsilon^2 g_{\mathrm{St}}^A$, using the actual link attachments. The overlap is an exact cone. Fixed local normalization factors are absorbed in the finitely many cap metrics. Denote the resulting surgery manifold by $\overline{X}_\epsilon$ and its metric

by $\bar g_\epsilon^{R,A}$. Its exterior and attachments are those of X_t; thus it is diffeomorphic to X_t, and the same Mayer–Vietoris sequence gives $b_1(\overline{X}_\epsilon) = 0$ and $b_2(\overline{X}_\epsilon) = s$.

Definition 4.2 (C^0 comparison)**.** A family of smooth metrics h_ϵ on the smoothing fibres satisfies condition (N) if, for every sufficiently small R and large A, there are diffeomorphisms $\Phi_\epsilon : \overline{X}_\epsilon \longrightarrow X_{\epsilon^3}$ such that

$$\limsup_{\epsilon\downarrow 0} \sup_{\overline{X}_\epsilon} |\Phi_\epsilon^* h_\epsilon - \bar g_\epsilon^{R,A}|_{\bar g_\epsilon^{R,A}} \le \delta(R,A), \qquad \delta(R,A) \longrightarrow 0 \quad (R\downarrow 0,\ A\to\infty). \tag{35}$$

The family limit is taken first. No derivatives or complex-structure compatibility of these diffeomorphisms are required.

Lemma 4.3. *The actual Ricci-flat family satisfies condition* (N)*. More precisely, under the collar-compatible identifications of Proposition 2.2, for some $b > 0$,*

$$\limsup_{\epsilon\downarrow 0} \sup_{X_t} |g_\epsilon - \bar g_\epsilon^{R,A}|_{\bar g_\epsilon^{R,A}} \le C(R^\mu + A^{-b}). \tag{36}$$

The family limit is taken with R, A fixed.

Proof. On $r \ge R$, the actual metric converges to g_0, and the central interpolation costs $O(R^\mu)$. On each fixed rescaled cap the actual metric converges to Stenzel, and the cap interpolation costs $O(A^{-a_1})$. In the intervening region $A\epsilon \le r \le R$, Proposition 2.2 and (14) give relative cone error $O(r^\mu) + O((\epsilon/r)^{a_0})$, bounded by $C(R^\mu + A^{-a_0})$. The corner coordinates agree at the two faces: $r = \rho_1$ and the cap radius is $r/\epsilon = 1/\rho_2$. Thus these estimates compare the same conic frames. Taking $b = \min(a_0, a_1)$ proves the assertion. The errors tending to zero with ϵ may depend on R, A, which is why the limits are ordered. □

5. Compactness and the number of small modes

All auxiliary central and cap metrics in this section are fixed. We work with $b = \dim V$, $s = \dim K$ and $m = b - s$ from Theorem 1.2.

5.1. Central harmonic forms and their domains.

Lemma 5.1. *For g_0^R, the smooth L^2 closed and coclosed one-forms vanish. The corresponding two-form space maps isomorphically to $H^2(M_0;\mathbb{R}) = V$. Every such two-form lies in the minimal graph domain of $d + d^*$.*

Proof. On an exact conical end write a closed and coclosed two-form as $u = a(r) + dr \wedge b(r)$. Link Hodge decomposition gives

$$a(r) = \sum_j c_j(r)\vartheta_j + d_L v(r), \qquad d_L^* v = 0,$$

where v is orthogonal to harmonic one-forms. The closedness equations imply $c_j' = 0$ and $b - v'$ closed. Coclosedness makes the latter harmonic. Since $H^1(T^{1,1}) = 0$, it vanishes, and

$$b = v', \qquad r^{-2}\Delta_L v - v'' - 3r^{-1}v' = 0.$$

On a positive coexact one-form eigenspace of link eigenvalue λ, the radial exponents of v are

$$k_\pm = -1 \pm \sqrt{1+\lambda}.$$

The negative root is not L^2 as a two-form. Therefore

$$u = \sum_j c_j\vartheta_j + dv_+, \quad |v_+| = O(r^{\kappa-1}), \quad |dv_+| = O(r^{\kappa-2}) \tag{37}$$

for some $\kappa > 0$, with differentiated estimates. Orthogonal link eigenfunction expansion and elliptic estimates on fixed-ratio annuli justify these bounds; higher positive roots decay faster on a smaller end. The harmonic link part admits the logarithmic cutoffs of Proposition 3.1, and the positive-root part admits ordinary fixed-ratio cutoffs with vanishing graph-norm error. This proves the minimal-domain assertion.

If u is globally exact, choose a smooth primitive w. Its harmonic link coefficients vanish. Since $H^1(T^{1,1}) = 0$, $w - v_+$ is locally exact. Subtracting differentials of cutoff potentials makes $w = v_+$ near all ends. This primitive is in L^2 and in the minimal domain of d: the squared cutoff error is $O(\rho^{2\kappa+2})$. The adjoint identity then gives $\|u\|^2 = \langle d_{\min} w, u\rangle = \langle w, d^*_{\min} u\rangle = 0$. Thus the map to ordinary cohomology is injective.

For surjectivity, use the closed minimal de Rham complex. Its reduced cohomology $\ker d_{\min}/\overline{\operatorname{im} d_{\min}}$ has harmonic representatives by Hilbert-space orthogonal decomposition; no closed-range assertion is needed for this reduced statement. The forms η_D for a basis of V, constructed in Section 3, remain closed minimal-domain forms for g_0^R, by norm equivalence. Their reduced classes are independent: pair them with compactly supported closed four-forms dual to that basis under $H^2(M_0) \times H^4_c(M_0)$. Those pairings are L^2-continuous and vanish on the closure of minimal exact forms. The resulting reduced harmonic representatives are smooth, closed and coclosed on the interior. Injectivity into the b-dimensional ordinary cohomology now gives the isomorphism.

For degree one, the cone calculation gives tangential harmonic link one-forms and differentials of positive-root harmonic functions; the radial solution $Cr^{-5}dr$ is not L^2. The positive scalar roots are $-2 + \sqrt{4+\lambda} > 0$. Since $H^1(M_0) = 0$, a harmonic one-form has a global primitive that is bounded up to a constant on each end. Its cutoff error has squared norm $O(\rho^4)$. The same adjoint argument makes the form zero. □

Let G_R be the harmonic Gram form on V for g_0^R, and let $\mathcal{M}_R$ be its quotient form on V/K. Since g_0^R differs relatively from g_0 by $O(R^\mu)$, their norms define the same closed minimal complex and reduced cohomology. Minimization over each class gives comparison factors $1 + O(R^\mu)$ for its harmonic norm. Polarization and finite-dimensional minimization over K therefore give

$$G_R \longrightarrow G, \qquad \mathcal{M}_R \longrightarrow \mathcal{M}. \tag{38}$$

In particular the generalized eigenvalues $\mu_j(R)$ of $(\mathcal{E}, \mathcal{M}_R)$ converge to μ_j. In rank one, if H_R, D_R are the central harmonic representatives, this quotient norm is

$$\alpha_R = D_R - \frac{\langle D_R, H_R\rangle}{\|H_R\|^2} H_R, \qquad \mathcal{M}_R([D],[D]) = \|\alpha_R\|^2. \tag{39}$$

The link coefficients of a representative remain its periods $r_i(D)$.

5.2. **Cap kernels and the critical degree.** We record the two facts needed to exclude cap mass in the spectral compactness theorem. The complete cap with metric g^A_{St} is a scattering manifold after $x = 1/r$. The scattering Hodge theorem [HHM04, Theorem 1A] identifies its L^2 harmonic forms below middle degree with $H^j_c(T^*S^3)$, above middle degree with $H^j(T^*S^3)$, and in middle degree with $\operatorname{im}(H^3_c \to H^3)$. All vanish. The first and last ordinary groups vanish by retraction onto S^3 and duality; the middle map is the Euler-class map of the cotangent bundle of S^3, which is zero because S^3 is parallelizable. This conclusion does not require the interpolated metric to be Kähler.

On a completed exact-cone cap, a *critical extended solution* is a smooth solution of $(d + d^*)u = 0$ with expansion $u = u_{\mathrm{crit}} + O(r^{-3-\kappa})$ at infinity for some $\kappa > 0$, with differentiated bounds. Here u_{crit} is homogeneous of pointwise order r^{-3}; a nonzero such term has logarithmically divergent squared L^2 norm. The critical coefficients of these global solutions, modulo L^2 solutions, form the trace space $I_{1/2}$ in [AT15, equation (8)]. This is the space used in its logarithmically normalized critical term. Throughout, physical degree means ordinary differential-form degree.

Lemma 5.2. *An actual critical extended solution of $d + d^*$ on a complete six-dimensional exact-cone-ended cap has its logarithmic L^2 leading term in physical degree three.*

Proof. Such a solution has $|u| = O(r^{-3})$ and an expansion in powers from the discrete tangential spectrum, with a positive gap after its critical term. First, $du = d^*u = 0$ separately, even if u has mixed degree. Choose a cutoff χ_T supported in $r < 2T$, equal one below T, with $|d\chi_T| \le C/T$. The shell mass is bounded, so

$$\|(d + d^*)(\chi_T u)\|^2 \le CT^{-2}.$$

For compactly supported forms on a boundaryless manifold this is $\|d(\chi_T u)\|^2 + \|d^*(\chi_T u)\|^2$. On every fixed compact set letting $T \to \infty$ proves the separate equations.

A homogeneous pure p-form of pointwise order r^{-3} has the form $u = r^{p-3}a + dr \wedge r^{p-4}b$. The leading closed/coclosed equations are

$$d_L a = 0, \quad (p-3)a = d_L b, \quad d_L^* b = 0, \quad d_L^* a + (p-3)b = 0.$$

If $p \neq 3$, they imply $\Delta_L b = -(p-3)^2 b$, hence $a = b = 0$. For $p = 3$ they state precisely that a and b are harmonic, yielding $a + (dr/r) \wedge b$. Thus every critical leading term lies in degree three. The argument applies to traces of actual complete-cap solutions, not to arbitrary formal endpoint eigenvectors of a boundary operator. The faster remainder is L^2 at infinity. After division by $\sqrt{|\log \epsilon|}$, its contribution to the critical term therefore tends to zero in L^2. □

5.3. **Several-cap norm compactness.**

Proposition 5.3. *Fix R, A. For any finite orthonormal collection of pure degree-p eigenforms on $(\overline{X}_\epsilon, \bar{g}_\epsilon^{R,A})$, $p = 1, 2$, with eigenvalues tending to zero, a common subsequence converges smoothly on fixed central compact subsets and strongly in central L^2 to closed and coclosed pure degree-p forms. There is no cap or critical mass loss, and the limiting Gram matrix is the identity.*

Proof. We specify the finite-cap consequence of [AT15]. Retain the closed Gauss–Bonnet extension D_W selected by its cap trace space W in equation (7), with Laplace operator $D_W^* D_W$. No identification of minimal and maximal domains, and no invariance of the boundary projectors under physical degree, is assumed. The pure-degree central kernels in Lemma 5.1 lie in the minimal graph domain and hence in every such selected domain.

Take the finite direct sum of the tangential operators A_i on the links. On each connected cap, Proposition 2 of the source gives the elliptic estimate; summing its squares and taking a finite maximum gives the estimate for their disjoint union. The Poisson extensions, Dirac–Neumann maps $T_i, T_{C,i}$ and spaces $W_i, I_{1/2,i}$ in its equations (5)–(8) are direct sums. Proposition 11 gives ellipticity and a finite-dimensional kernel for $T_i + T_{C,i}$, and closedness of the relevant restricted image of T_i. No condition couples the boundary values of different cap components.

The transmission formulas (9)–(15) and the radial estimates of Lemmas 5–8 apply on each collar. In every bounded spectral interval the finite union of link spectra has only finitely many eigenvalues, so the required positive separations can be chosen uniformly over these links. Propositions 9–12 give the cap and critical limits componentwise. Propositions 13–15 and Lemma 16 estimate sums of disjoint radial integrals; their tail and almost-orthogonality estimates therefore sum. The exterior elliptic estimate of Lemma 4 is unchanged. A common subsequence suffices for finitely many caps and vectors. Thus Corollary 17, equations (24)–(25), and Lemma 18, equation (26), give on the central region

$$u_\epsilon^a = \Phi_\epsilon^a + V_\epsilon^a + C_\epsilon^a + o_{L^2}(1), \qquad \Phi_\epsilon^a \longrightarrow u_0^a \quad \text{strongly}, \tag{40}$$

where V is the complete-cap harmonic tail and C is the logarithmically normalized critical tail. If v_i^a is the full complete-cap harmonic limit and σ_i^a its critical trace, the simultaneous decomposition and almost orthogonality give, with the left pairing on the full glued manifold,

$$\lim \langle u_\epsilon^a, u_\epsilon^b \rangle = \langle u_0^a, u_0^b \rangle + \sum_i \langle v_i^a, v_i^b \rangle + \sum_i \langle \sigma_i^a, \sigma_i^b \rangle. \tag{41}$$

The full norm of v_i^a includes its cap core. Polarization of the simultaneous decomposition gives the cross terms, rather than inferring them from separate unit-norm statements.

The critical coefficient first occurs in $\ker(A_i - 1/2)$. The first step of Proposition 12 places it in the kernel of the obstruction from Proposition 11. That proposition supplies a faster trace η_i with $T_i(\sigma_i + \eta_i) = 0$. It is therefore the trace of a global complete-cap $d + d^*$ solution with no slower terms, namely the leading-coefficient image of an element of the critical trace quotient $I_{1/2,i}$ in equation (8). This extension property is essential when applying Lemma 5.2. In particular, the extra formal endpoint channel in Remark 20 of the source, from coexact link two-forms with eigenvalue one, is not presumed absent; an actual critical trace in that channel must satisfy the same global closedness and coclosedness argument.

The complete-cap L^2 kernels vanish by the calculation above, so V and all v_i^a vanish. Since $\|du_\epsilon^a\|^2 + \|d^* u_\epsilon^a\|^2 \to 0$, the central limits are separately closed and coclosed, and have the original physical degree. Lemma 5.2 puts the leading part of C in degree three, with its faster remainder tending to zero after logarithmic normalization. Apply degree-three projection to the actual identity (40). The original form and limiting central form project to zero. Hence $C \to 0$ in norm. Equation (41) now proves the result. This projection has not been commuted through a boundary spectral projector.

The source proof has thus been used only through estimates on the fixed exterior and finitely many individual cap problems. Connectedness of their union is unnecessary. The topology of the resulting manifold was computed separately from its actual attachments in Section 4. □

5.4. **Counting the modes.**

Proposition 5.4. *For fixed R, A, the conic models have exactly $m = b - s$ positive degree-two eigenvalues tending to zero, and uniform positive lower bounds for the degree-one spectrum and the remaining positive degree-two spectrum.*

Proof. Proposition 5.3 and the central dimensions zero and b give the upper counts, including topological zeros. Conversely, approximate an orthonormal basis of all b central harmonic two-forms by compactly supported pure two-forms in the minimal graph norm. Their energies tend to zero and their Gram matrix tends to the identity. For each fixed approximation extend by zero across the caps; the model equals the central metric on its support for small ϵ. Min–max, first in ϵ and then in the approximation, gives at least b zero-limit eigenvalues. Exactly

s are identically zero by the exterior attachment calculation. Failure of either complementary positive gap would produce too many orthonormal central limits by the same compactness proposition. □

6. The ordered spectral law and its transfer

6.1. **The surviving model harmonic space.** Fix R, A. For $k \in K$, let η_k^R be its central harmonic representative. Its harmonic link coefficients vanish. The exact-cone expansion (37) gives an end primitive β_k of size $O(r^{\nu-1})$, for some $\nu > 0$, with differentiated bounds. If χ_ρ equals one near each tip and zero above 2ρ, then replacing the form on its ends by

$$\gamma_{k,\rho} = \eta_k^R - d(\chi_\rho \beta_k) \tag{42}$$

gives a compactly supported closed representative of k converging in L^2 to η_k^R. The squared tail and cutoff error is $O(\rho^{2\nu+2})$.

Let $h_\epsilon^R(k)$ be the harmonic representative on the model of the corresponding surviving class. Extend (42) by zero over the caps. Minimization first in the fixed model and then $\rho \downarrow 0$ gives

$$\limsup_{\epsilon \to 0} \|h_\epsilon^R(k)\|^2 \le G_R(k,k).$$

For nonzero k, a fixed compactly supported closed dual four-form bounds these norms below. Proposition 5.3 at eigenvalue zero gives a central limit without lost norm. Compactly supported dual pairings identify its class as k, and Lemma 5.1 identifies the limit as η_k^R. Consequently the full harmonic Gram matrix on K converges to $G_R|_K$. All elliptic convergence used here is on the fixed central model metric. No assertion about derivatives of harmonic representatives for arbitrary metrics satisfying (N) is needed.

6.2. **Upper and lower bounds on every model subspace.** Let $D \in K^{\perp_{G_R}}$. On end i write

$$\eta_D^R = r_i(D)\vartheta_i + dv_{+,D}, \qquad |dv_{+,D}| = O(r^{\kappa-2}), \quad \kappa > 0.$$

Take the Lipschitz cutoff equal to zero below $2A\epsilon$, equal to one above R, and equal to

$$\chi_\epsilon(r) = \frac{\log(r/(2A\epsilon))}{\log(R/(2A\epsilon))}$$

between these radii. The form $T_\epsilon D = \chi_\epsilon \eta_D^R$, extended by zero over the caps, lies in the model Hodge form domain. The metric is exactly conical where the cutoff varies. Closedness and coclosedness of η_D^R give $dT_\epsilon D = d\chi_\epsilon \wedge \eta_D^R$ and $d^* T_\epsilon D = -\iota_{\nabla \chi_\epsilon} \eta_D^R$. The radial contraction of ϑ_i vanishes, and its tangential pairing with $d_L v_{+,D}$ is zero. With $L_\epsilon = \log(1/\epsilon)$ and $c = 2\pi/3$, it follows that

$$\bar{q}_\epsilon(T_\epsilon D) = \frac{c\mathcal{E}([D],[D])}{L_\epsilon} + O_{R,A}(L_\epsilon^{-2})\|D\|^2, \tag{43}$$

$$\langle T_\epsilon D, T_\epsilon F\rangle = \mathcal{M}_R([D],[F]) + O_{R,A}(L_\epsilon^{-1})\|D\|\|F\|. \tag{44}$$

Here any fixed norm on V may be used. The remainder energy and mass error are bounded, respectively, by

$$CL_\epsilon^{-2}\int_0^R r^{2\kappa-1}\,dr, \qquad CL_\epsilon^{-1}\int_0^R r(1+|\log r|)\,dr + O(\epsilon^2).$$

Polarization gives the bilinear estimates, uniformly on this fixed finite-dimensional space.

Subtract the full model harmonic projection. Its norm is $o(1)$ by the preceding harmonic convergence and $G_R(D,K)=0$; the energy is unchanged. Min–max on the first j generalized eigendirections of $(\mathcal{E},\mathcal{M}_R)$ gives

$$\limsup_{\epsilon\to 0} L_\epsilon \bar\lambda_j(\epsilon;R,A) \le c\mu_j(R), \qquad 1\le j\le m. \tag{45}$$

The $o(1)$ projection estimate suffices for this leading limit and does not assert a second-order eigenvalue remainder.

For the lower bound choose a sequence on which $L_\epsilon\bar\lambda_j$ tends to its lower limit ℓ; the upper bound makes it finite. The first j orthonormal model eigenforms have a common subsequence with orthonormal central limits $v_1,\dots,v_j\in K^{\perp_{G_R}}$, by Proposition 5.3 and convergence of the model kernel. For $z\in\mathbb{R}^j$ their combination $u_\epsilon(z)$ has

$$\bar q_\epsilon(u_\epsilon(z)) \le \bar\lambda_j|z|^2, \qquad v(z)=\sum_{a=1}^{j} z_a v_a.$$

At the fixed sections $r_*=R/2$, its harmonic boundary coefficients converge smoothly to $r_i(v(z))$. Each inner region is a scaled fixed cap with an exact conical end. Lemma 4.1 gives

$$\bar q_\epsilon(u_\epsilon(z)) \ge \sum_i \frac{c|B_{i,\epsilon}(z)|^2}{\log(r_*/\epsilon)+C_{i,A}}.$$

The regions are disjoint. Multiplying by L_ϵ and taking the limit yields, simultaneously for the finite basis and hence every z,

$$c\mathcal{E}(v(z),v(z)) \le \ell|z|^2 = \ell\mathcal{M}_R(v(z),v(z)).$$

The largest generalized Rayleigh quotient on this j-dimensional span is at least $\mu_j(R)$. Together with (45), this proves

$$\lim_{\epsilon\to 0} L_\epsilon\bar\lambda_j(\epsilon;R,A) = c\mu_j(R), \qquad 1\le j\le m. \tag{46}$$

The constants $C_{i,A}$ require no uniform bound as A varies, since the family limit has been taken first.

6.3. **Transfer under C^0 comparison.**

Lemma 6.1 (Dodziuk comparison)**.** *Let g,h be smooth metrics on a closed n-manifold and suppose $(1-\delta)g \le h \le (1+\delta)g$, with $0<\delta<1/2$. For every ordered positive Hodge eigenvalue in degree p,*

$$K(\delta)^{-1}\lambda^+_{p,j}(g) \le \lambda^+_{p,j}(h) \le K(\delta)\lambda^+_{p,j}(g), \qquad K(\delta)=1+O_n(\delta).$$

No derivatives of the metric comparison are required.

Proof. This is the closed-manifold form of [Dod82, Proposition 3.3]. We recall the variational argument to specify the metric dependence. The squared L^2 norms of p-forms satisfy

$$a_p\|u\|_g^2 \le \|u\|_h^2 \le b_p\|u\|_g^2, \quad a_p=(1-\delta)^{n/2}(1+\delta)^{-p}, \quad b_p=(1+\delta)^{n/2}(1-\delta)^{-p}.$$

The distributional differential d and its graph domain are unchanged under equivalent norms. Set $Z^p=\ker d$ and give $\operatorname{Dom} d/Z^p$ the quotient norm $\|[u]\|^2_{g,q}=\inf_{z\in Z^p}\|u+z\|_g^2$. The same constants compare these quotient norms. The coexact positive p-eigenvalues are the squared singular values of d on this quotient, with min–max quotient $\|du\|^2_{p+1,g}/\|[u]\|^2_{g,q}$. Its comparison factors are a_{p+1}/b_p and b_{p+1}/a_p. The positive exact p-spectrum equals the coexact $(p-1)$-spectrum. The full positive spectrum is the ordered union of these two lists,

with multiplicity; the same common multiplicative bounds hold for their ordered union. This proves the lemma and avoids comparing codifferentials pointwise. □

Theorem 6.2 (Spectral consequence of condition (N))**.** *Under the geometric hypotheses of Theorem 1.2, let h_ϵ be any smooth metric family satisfying condition* (N) *relative to the central metric g_0. Then exactly m positive degree-two eigenvalues tend to zero, with*

$$\lim_{\epsilon\downarrow 0} \log(1/\epsilon)\lambda^+_{2,j}(h_\epsilon) = \frac{2\pi}{3}\mu_j, \qquad 1 \le j \le m.$$

There are uniform positive complementary gaps in degrees one and two. If these metrics are Ricci-flat Kähler with trivial canonical bundle, the small spectral space is coexact and primitive of real type $(1,1)$.

Proof. Fix R, A. For every $\eta > 0$ and all sufficiently small ϵ, condition (N) bounds the relative metric error by $\delta(R,A) + \eta$. Take this sum small enough for Lemma 6.1. Its multiplicative factor $K(\delta+\eta)$ bounds each ordered positive eigenvalue above and below. Using (46), then letting $\eta \downarrow 0$, $R \downarrow 0$ and $A \to \infty$, proves the coefficient because $\mu_j(R) \to \mu_j$. One fixed admissible pair R, A transfers the model complementary gaps by a fixed positive factor. No gap uniform in the auxiliary radii is needed. The Kähler assertion follows from the next lemma. □

Lemma 6.3. *On a compact Ricci-flat Kähler threefold with trivial canonical bundle, every real two-form eigenvector with positive eigenvalue below the first positive one-form eigenvalue is coexact and primitive of type* $(1,1)$.

Proof. A nowhere-zero holomorphic three-form Ω is parallel. Using the complex-bilinear metric to raise indices, the map

$$T : \Lambda^{0,1} \longrightarrow \Lambda^{2,0}, \qquad T\beta = \iota_{\beta^\sharp}\Omega$$

is a complex-linear parallel bundle isomorphism, equivariant for $SU(3)$ and hence for the actual holonomy group. It intertwines the Hodge Laplacians: both the rough term and the curvature term $q(R)$ in the Weitzenböck formula are natural under this holonomy intertwiner [SW19, §§2.4, 2.6, Definition 2.4 and Example 1]. Explicitly, $q(R) = \sum_\alpha \rho(X_\alpha)\rho(R(X_\alpha))$ in an orthonormal holonomy-algebra frame. Thus a nonzero $(2,0)$ eigencomponent would give a nonzero $(0,1)$ eigenform of the same eigenvalue via T^{-1}, contradicting the one-form gap. Reality removes the conjugate component.

Kähler commutation makes Λu a scalar eigenfunction of the same positive eigenvalue. If nonzero, its differential gives the same contradiction. Similarly d^*u must vanish. Hence $\Lambda u = 0$ and $u = \lambda^{-1} d^* du$. The argument applies to the entire eigenspace and requires no simplicity hypothesis. □

Lemma 4.3 establishes condition (N) for the actual polarized Ricci-flat metrics. Substituting $\epsilon = t^{1/3}$ in Theorem 6.2 proves Theorem 1.2. A simple μ_j is separated from its neighbours; the ordered laws and complete count imply eventual simplicity. In rank one the quotient mass is (22), so the same proof gives all assertions of Theorem 1.1 except its sharper upper remainder.

6.4. **The differentiated rank-one remainder.** The remaining assertion uses the stronger full-central estimate (17), including its first derivative. Impose $b_2(Y) = 2$, $b_2(X_t) = 1$ for the rest of this subsection. We now use the global form α of Proposition 3.1. Put $L_\epsilon = \log(1/\epsilon)$.

On end i choose a cutoff $\chi_{i,\epsilon}$ vanishing below $A_i\epsilon$, equal to one above a fixed small R_i, and logarithmic between these radii:

$$\chi_{i,\epsilon}(r) = \frac{\log(r/(A_i\epsilon))}{\log(R_i/(A_i\epsilon))}.$$

Round its endpoints smoothly on fixed multiplicative annuli, making it flat at the inner endpoint. The rounding changes energy by $O(L_\epsilon^{-2})$. Denote the denominator by $L_i = L_\epsilon + O(1)$. Transport $\chi_\epsilon\alpha$ using F_ϵ and extend by zero across the cores, obtaining a smooth real two-form u_ϵ on X_t.

Proposition 6.4. *For the actual Ricci-flat metrics,*

$$\lambda_2^+(\epsilon) \le \frac{c\sum_i b_i^2}{ML_\epsilon} + O(L_\epsilon^{-2}), \qquad c = 2\pi/3. \tag{47}$$

Proof. Pullback commutes with d, so du_ϵ is the transport of $d\chi_\epsilon \wedge \alpha$. On end i its leading energy is $cb_i^2 \int r|\chi_i'|^2 dr = cb_i^2/L_i + O(L_\epsilon^{-2})$. The errors in the form, central metric and actual metric carry respectively positive factors r^ν, r^μ and $(\epsilon/r)^{a_0}$. Their integrals against dr/r are bounded, and each has prefactor $O(L_\epsilon^{-2})$. Thus

$$\|du_\epsilon\|^2 = \frac{c\sum_i b_i^2}{L_\epsilon} + O(L_\epsilon^{-2}). \tag{48}$$

The actual codifferential requires a separate estimate. Proposition 2.2 and the differentiated $O(r^{-2})$ bound for α give

$$|(d_{g_\epsilon}^* - d_{g_0}^*)\alpha| \le C(\epsilon/r)^{a_0} r^{-3}.$$

Since $d_{g_0}^*\alpha = 0$, the contribution multiplied by χ_i has squared norm at most

$$C\int_{A_i\epsilon}^{R_i} \chi_i^2(\epsilon/r)^{2a_0}\frac{dr}{r} \le \frac{C}{L_\epsilon^2}\int_{A_i}^{\infty}(1+\log(s/A_i))^2 s^{-1-2a_0}ds = O(L_\epsilon^{-2}).$$

The product-rule contraction obeys

$$|\iota_{\nabla_{g_\epsilon}\chi_i}\alpha| \le C|\chi_i'|\big(r^{\nu-2} + (\epsilon/r)^{a_0}r^{-2}\big),$$

because the radial cone contraction of ϑ_i vanishes. Its squared norm has bounded radial integrals $\int r^{2\nu-1}dr$ and $\int(\epsilon/r)^{2a_0}dr/r$, again with prefactor $O(L_\epsilon^{-2})$. On the fixed exterior the squared error is $O(\epsilon^{2a_0})$. Combining with (48),

$$q_\epsilon(u_\epsilon) := \|du_\epsilon\|^2 + \|d^*u_\epsilon\|^2 = \frac{c\sum_i b_i^2}{L_\epsilon} + O(L_\epsilon^{-2}). \tag{49}$$

The mass density of α is $O(r\,dr)$, and the removed inner mass is $O(\epsilon^2)$. On the remainder, $|1-\chi_i| \le C(1+|\log r|)/L_\epsilon$, which is integrable against that density. The actual metric error integrates to $O(\epsilon^{\min(a_0,2)}(1+|\log\epsilon|))$. Therefore

$$\|u_\epsilon\|^2 = M + O(L_\epsilon^{-1}).$$

The Kähler-form comparison and pointwise central primitivity similarly give $\langle u_\epsilon, \omega_\epsilon\rangle = O(L_\epsilon^{-1})$. Subtract its orthogonal projection onto ω_ϵ. Since $\|\omega_\epsilon\|^2 = v/2$, this changes the squared norm by $O(L_\epsilon^{-2})$ and leaves q_ϵ unchanged. It removes the whole harmonic kernel because $b_2(X_t) = 1$. The Rayleigh principle then proves (47). □

Since $\log(1/\epsilon) = |\log t|/3$, Proposition 6.4 gives (5) and completes the proof of Theorem 1.1. This estimate is separate from the C^0 transfer and has only the stated one-sided conclusion.

7. The split-quintic coefficient

The sixteen-node split quintic provides the rank-one application. We specify the family and calculate its coefficient here. Appendix A computes the integral rings, attachments, specialization maps and monodromy of the same transition; Appendix B treats its tangent sheaves.

7.1. **The genuine global family.** Choose independent linear forms l_1, l_2 and general quartics q_1, q_2 on $\mathbb{P}^4$, and set

$$\begin{aligned} F_0 &= l_1 q_2 - l_2 q_1, \qquad X_0 = \{F_0 = 0\}, \\ Y &= \{u l_1 + v l_2 = 0,\ u q_1 + v q_2 = 0\} \subset \mathbb{P}^4 \times \mathbb{P}^1, \\ X_t &= \{F_0 + tG = 0\}. \end{aligned} \tag{50}$$

Here G is a quintic nonzero at the sixteen points of $q_1 = q_2 = 0$ in $\Pi = \{l_1 = l_2 = 0\}$. The precise open conditions are transversality of the two restricted quartics, smoothness of X_0 off Π, and smoothness of Y.

Proposition 7.1. *These conditions hold on a nonempty open set. The projection $q : Y \to X_0$ is a projective small crepant resolution of exactly sixteen ordinary nodes. After shrinking the parameter disc, (50) is a flat polarized family with trivial relative canonical bundle and all noncentral fibres smooth. Every node is smoothed to first order. Both Y and X_t are simply connected.*

Proof. The linear system $l_1 q_2 - l_2 q_1$ is the full degree-five ideal of Π and is basepoint free off Π, so Bertini gives smoothness there. On Π its differential is $q_2 dl_1 - q_1 dl_2$. General plane quartics meet transversely in sixteen points by Bézout. At each intersection, (l_1, l_2, q_1, q_2) are analytic coordinates: the first two are normal to Π and the last two are independent along it. The local equation is the quadratic determinant, an ordinary double point.

The incidence divisor $u l_1 + v l_2 = 0$ is smooth, including over Π, where its normal differential $u\, dl_1 + v\, dl_2$ is nonzero. The second sections $u q_1 + v q_2$ form a basepoint-free system there. Bertini gives smooth Y. Projection has a unique kernel line at rank-one matrices and a whole $\mathbb{P}^1$ at the sixteen rank-zero matrices. The determinant coordinates identify its local resolution with $\mathcal{O}_{\mathbb{P}^1}(-1)^{\oplus 2}$. Adjunction for the two ample divisor classes $H + L$ and $4H + L$ gives $K_Y = 0$.

The relative hypersurface is flat over a sufficiently small disc. The condition $G(p_i) \neq 0$ gives a simple local smoothing; equivalently the holomorphic Morse lemma applies to F_0/G. Outside the node neighborhoods, compactness prevents a sequence of critical points from approaching the smooth central exterior. Thus all sufficiently small nonzero fibres are smooth. Adjunction gives trivial relative canonical bundle. Ehresmann's theorem with transverse boundary identifies the exteriors. Finally integral Lefschetz, twice for Y and once for the smooth quintic, gives simple connectivity and the integral degree-two markings used below. □

7.2. **The spectral coefficient and the allowed parameters.** Integral Lefschetz gives $b_2(Y) = 2$ and $b_2(X_t) = 1$, with generators H, L and H, respectively. The fundamental class of Y in $\mathbb{P}^4 \times \mathbb{P}^1$ is $(H + L)(4H + L)$, so

$$\int_Y H^3 = 5, \qquad \int_Y H^2 L = 4, \qquad L^2 = 0. \tag{51}$$

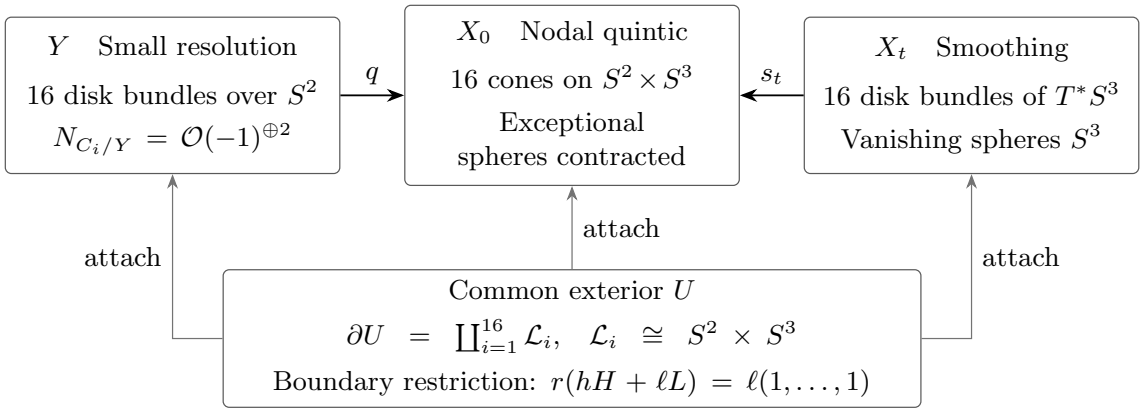


FIGURE 1. The common exterior and its three fillings. The map q is algebraic; s_t is the continuous Milnor collapse along the chosen ray. The resolution and smoothing cores have real dimensions two and three. The same boundary restriction gives the vanishing-cycle relation and the periods in the spectral coefficient.

Every exceptional curve has $(H \cdot C_i, L \cdot C_i) = (0, 1)$. Proposition 7.1 therefore verifies the remaining hypotheses of Theorem 1.1. Taking $D = L$ gives

$$M = \frac{4^2}{5} - 0 = \frac{16}{5}, \qquad \frac{2\pi \sum_{i=1}^{16} (L \cdot C_i)^2}{M} = 10\pi,$$

which proves Corollary 1.3 and the spectral assertion of Theorem A.1. By Proposition 3.3, the vector $(L \cdot C_i)_i = \mathbf{1}$ in this calculation is the same vector that generates the kernel of the vanishing-cycle map. The denominator is the polarized primitive norm from the same integral ring. It is not an independent local coefficient fitted to the spectrum.

The family has one exceptional-area direction. Arbitrary independent node parameters are not implied by the number of nodes. Already at first order, evaluation of quintics at the sixteen planar points has rank fifteen: the degree-five Hilbert function of a plane $(4, 4)$ complete intersection is $\binom{7}{2} - 2\binom{3}{2} = 15$. Cayley–Bacharach makes the unique evaluation relation nonzero at every point. The actual pencil selects a compatible vector nonzero at all nodes. This is sufficient for the fixed-ray theorem and does not supply uniformity for arbitrary independent smoothing rates.

8. THE EIGHT-NODE TRANSITION AND ITS TWO SMALL MODES

We apply Theorem 1.2 to the compact transition constructed in stability–degeneration Paper 2, [W26e, Appendix A, Proposition A.1]. This uses its geometric realization and exceptional-curve data as premises. The additional deformation argument below supplies a polarized smoothing with the precise first-order and canonical-bundle properties required by the analytic theorem. This family has dual-edge length four at its nontriangular toric faces; it is outside the length-one admissible class of CY VI [W26k, Definition 2.1].

8.1. **A polarized simple smoothing of the earlier example.** Let $B = \mathbb{P}^1 \times \mathbb{P}^1$, $L = \mathcal{O}_B(1, -1)$, and

$$Z = \mathbb{P}_B(\mathcal{O} \oplus L) \times_B \mathbb{P}_B(\mathcal{O} \oplus L). \tag{52}$$

Use the quotient convention for the tautological classes ξ_1, ξ_2, and write f, g for the base hyperplane classes. Let $\bar{Z}$ be the anticanonical model, $Y \subset Z$ a general anticanonical hypersurface chosen simultaneously nondegenerate for its toric polytope, and $q : Y \to X_0 \subset \bar{Z}$ the induced contraction. Set

$$\mathcal{L}_0 = (-K_{\bar{Z}})|_{X_0}, \qquad H = c_1(q^*\mathcal{L}_0) = (4g + 2\xi_1 + 2\xi_2)|_Y. \tag{53}$$

Proposition 8.1. *The threefold X_0 has eight ordinary double points and the displayed map is a smooth simply connected projective small crepant resolution. There is a proper flat projective smoothing over a disc, with trivial relative canonical bundle and a relatively ample line bundle restricting to the exact $\mathcal{L}_0$, such that every local smoothing function has nonzero first derivative. The smooth fibres are simply connected, and the second Betti numbers of Y and X_t are four and two.*

Proof. The cited construction proves that $\bar{Z}$ has two disjoint singular projective lines with transverse ordinary-node singularities. A general anticanonical section meets each in four simple points. Its projective crepant toric refinement restricts to q, and the resulting smooth Y has Hodge pair $(4, 60)$, trivial canonical bundle and $H^1(\mathcal{O}_Y) = H^2(\mathcal{O}_Y) = 0$. Rationality of the nodes transfers both vanishings to X_0. The model $\bar{Z}$ is Gorenstein Fano, so $\mathcal{L}_0$ is ample and Cartier. The canonical-class calculation in the quotient convention gives (53), and Cartier adjunction gives $\omega_{X_0} \simeq \mathcal{O}_{X_0}$.

The four-plus-four multiplicity and the curve periods can also be read directly from the bundle construction. Let $S_f, S_g \simeq B$ be the sections for which both quotients are respectively onto $\mathcal{O}$ and onto L. Their normal bundles and anticanonical restrictions are

$$N_{S_f/Z} = (L^{-1})^{\oplus 2}, \quad N_{S_g/Z} = L^{\oplus 2}, \qquad (-K_Z)|_{S_f} = \mathcal{O}(0,4), \quad (-K_Z)|_{S_g} = \mathcal{O}(4,0).$$

The restrictions of global sections are complete linear systems: the all-$\mathcal{O}$ and all-L summands in

$$\pi_*\mathcal{O}_Z(-K_Z) = \mathcal{O}(0,4) \otimes \mathrm{Sym}^2(\mathcal{O} \oplus L) \otimes \mathrm{Sym}^2(\mathcal{O} \oplus L)$$

have independent coefficients. A general section therefore meets each S in four distinct fibres of its contracted ruling. For one such fibre C, the sequence with $N_{C/S} = \mathcal{O}$ and $N_{S/Z}|_C = \mathcal{O}(-1)^{\oplus 2}$ splits, since $H^1(\mathbb{P}^1, \mathcal{O}(1)) = 0$. The normal derivative of the anticanonical equation is an isomorphism on the trivial summand because the corresponding quartic zero is simple. Hence $N_{C/Y} = \mathcal{O}(-1)^{\oplus 2}$. The divisor degrees in the basis (f, g, ξ_1, ξ_2) are

$$\Gamma_f = (1, 0, 0, 0), \qquad \Gamma_g = (0, 1, -1, -1). \tag{54}$$

The imported global contraction calculation excludes further exceptional loci or singularities, and its Picard-rank computation shows that these four divisor classes span $H^2(Y; \mathbb{Q})$. Thus the displayed degrees determine the rational exceptional homology classes. Its full rational relation space is given by separate sum-zero conditions on these two groups of four. In particular $(1, 1, 1, -3; 1, 1, 1, -3)$ has full support. The first-order criterion of [RT09, Theorem 1.2 and Section 3, (3.2)] gives a deformation tangent nonzero in every local node-smoothing space. Theorem 1.6 of [Fri26] makes this tangent unobstructed: X_0 is a compact Gorenstein Calabi–Yau threefold, its isolated rational lci singularities are Du Bois, its projective resolution satisfies the $\partial\bar{\partial}$ lemma, and $H^1(\mathcal{O}_{X_0}) = 0$.

We verify projective effectivity and the exact polarization. Choose N so that $\mathcal{L}_0^N$ is very ample and $H^1(\mathcal{L}_0^N) = 0$. Over a small Artin extension, the vanishing of $H^2(\mathcal{O}_{X_0})$ lifts the embedding line bundle, and $H^1(\mathcal{L}_0^N) = 0$ lifts its coordinate sections. The lifted morphism remains a closed immersion over the nilpotent thickening. Thus embedded deformations map formally smoothly to abstract deformations; equivalently the restricted Euler sequence gives

$H^1(T_{\mathbb{P}^M}|_{X_0}) = 0$. The corresponding Hilbert-scheme germ is smooth. Retain a lift of the selected tangent and shrink to a neighbourhood S with integral fibres.

Perform the Picard construction over this scheme S before choosing the analytic disc. After an étale neighbourhood, the universal family has a section through a smooth point of X_0 [Stacks, Tag 055S]. Its proper integral fibres give the universal identity $\mathcal{O}_T \simeq p_{T*}\mathcal{O}$ under base change, where p is the universal family projection. The relative Picard scheme exists, and the section supplies a universal Poincaré line bundle, normalized along that section [Kle05, Theorems 2.5(2), 4.8; Exercises 3.11, 4.3 and their Appendix A solutions]. Since $H^2(\mathcal{O}_{X_0}) = 0$, the Picard morphism is smooth at $[\mathcal{L}_0]$ [Kle05, Proposition 5.19]. A further étale neighbourhood gives a section through this point; pulling back the Poincaré bundle extends precisely $\mathcal{L}_0$. Ampleness is open. Both base changes have isomorphic tangent maps at the chosen complex points. A holomorphic disc in the resulting smooth base with the prescribed tangent therefore gives a relatively polarized projective family and retains every nonzero local derivative. No ramified reparametrization or replacement of H by NH occurs.

At a node the local equation is $\sum z_j^2 = \tau_i(t)$ with $\tau_i'(0) \neq 0$, so the total space is smooth. Properness allows the disc to be shrunk so that the entire total space is smooth and the family is Gorenstein. Its relative dualizing line bundle restricts to $\omega_{X_0} \simeq \mathcal{O}_{X_0}$. Since $H^1(\omega_{X_0}) = 0$, cohomology and base change lifts a central nowhere-zero generator to a relative section. Its zero set has proper image missing the origin. Shrinking the disc removes that image and trivializes the actual relative canonical bundle.

For simple connectivity, apply [BK06, Corollary 1.9] to the nondegenerate toric hypersurface and the smooth projective crepant subdivision in the construction. Its lattice quotient is zero: the relevant dual-polytope vertices include the lattice basis e_1, e_2, f_0, g_0. Thus $\pi_1(Y) = 0$. The common exterior has simply connected links $S^2 \times S^3$, and the two kinds of filling retract onto S^2 and S^3. Van Kampen gives $\pi_1(X_t) = \pi_1(Y) = 0$. Finally the two independent exceptional classes give $b_2(X_t) = 4 - 2 = 2$, as also computed in the earlier proposition. □

These properties verify the general smoothability hypothesis of Hein–Sun and the smoothing hypotheses of Benabida used in Section 2. Locally write $\tau_i(t) = tu_i(t)$ with $u_i(0) \neq 0$; a holomorphic square root of u_i changes the coordinates to the model $\sum z_j^2 = t$ over the same base. The Stenzel metric supplies the exact affine Kähler model. The finite-node construction has fixed positive leading scale ratios, and neither metric theorem imposes $b_2(X_t) = 1$. In particular the actual metrics in class H satisfy the comparisons used in Theorem 1.2.

8.2. **The quotient mass and the two coefficients.** The earlier construction calculates

$$\left(\int_Y D_iD_jH\right)_{i,j} = \begin{pmatrix} 0 & 8 & 4 & 4 \\ 8 & 0 & 12 & 12 \\ 4 & 12 & -8 & 0 \\ 4 & 12 & 0 & -8 \end{pmatrix}, \qquad (D_i) = (f, g, \xi_1, \xi_2). \tag{55}$$

It follows directly from $f^2 = g^2 = 0$, $\xi_i^2 = (f - g)\xi_i$ and $\int_Z fg\xi_1\xi_2 = 1$, by integrating $D_iD_j(-K_Z)^2$; the same calculation gives $H^3 = 320$. The restriction kernel has basis

$$k_1 = g + \xi_1, \qquad k_2 = g + \xi_2, \qquad H = 2k_1 + 2k_2.$$

Choose quotient lifts f, g. By (54), their period vectors have disjoint four-entry unit supports. The energy matrix is $\mathsf{E} = 4I_2$. The intersection blocks are

$$B_{KK} = \begin{pmatrix} 16 & 24 \\ 24 & 16 \end{pmatrix}, \quad B_{DK} = \begin{pmatrix} 12 & 12 \\ 12 & 12 \end{pmatrix}, \quad B_{DD} = \begin{pmatrix} 0 & 8 \\ 8 & 0 \end{pmatrix}.$$

Consequently (28) gives

$$\mathsf{M} = \frac{4}{5}\begin{pmatrix} 9 & -1 \\ -1 & 9 \end{pmatrix}. \tag{56}$$

The orthogonal primitive lifts

$$D_- = f - g, \qquad D_+ = f + g - \frac{3}{5}(k_1 + k_2)$$

have squared masses 16 and 64/5, respectively, and both have squared period norm 8. Thus the generalized eigenvalues of (E, M) are 1/2 and 5/8. Theorem 1.2 proves (8), including the count, eventual simplicity, type and gaps, and completes the proof of Corollary 1.4.

Only rational divisor data are used here. No new integral basis, saturation or full cohomology-ring assertion is made for this family. The application uses the earlier compact construction to derive new information about its actual metric operators. It preserves the distinction established there between tangent-stability walls in the ambient fourfold, toric contraction circuits and Calabi–Yau cycle relations.

9. Three small modes on a four-node toric transition

We now use the original-side nodal partial resolution X_9' constructed in CY VI [W26k, Section 11.1 and Appendix G.1]. That paper supplies its four nodes, a projective small resolution and the integral exceptional-curve relation. The deformation results of CY V supply a prescribed first-order smoothing tangent and its integrability. We specify an exact polarization, verify the remaining family hypotheses, and compute three spectral coefficients from the full quotient harmonic norm. This connects the earlier geometric construction to Theorem 1.2.

The notation distinguishes three spaces. The original X_9 of CY V has two nodes and a degree-seven del Pezzo cone; it does not satisfy our nodal hypotheses. Its partial resolution X_9' has four nodes. The four-node mirror studied elsewhere in CY VI is a different threefold. Only X_9' is used in this section.

9.1. **The earlier nodal model and its polarization.** The eleven rays $v_0, \dots, v_{10}$ of the original-side ambient fan are the columns of

$$\begin{pmatrix} 1 & 0 & -1 & 0 & 0 & -4 & -4 & 3 & 2 & -1 & -2 \\ 0 & 1 & -1 & 0 & 0 & -2 & -2 & 1 & 1 & 0 & -1 \\ 0 & 0 & 0 & 1 & 0 & -1 & 0 & 1 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & -1 & 1 & 1 & 0 & 0 \end{pmatrix}. \tag{57}$$

The first nine are the vertices of the reflexive polytope Δ_9; v_9 is interior to a facet and v_{10} is the pentagon centre. Use the source's integral support vector

$$h = (2,2,2,3,3,3,3,2,2,1,2), \qquad P_h = \{u \in \mathbb{R}^4 : h_i + \langle u, v_i \rangle \geq 0 \text{ for all } i\}. \tag{58}$$

Its normal fan is the source fan Σ' with twenty-one maximal cones. The vertices of P_h are

$$\begin{array}{ccc} (-2,-2,-3,15) & (-2,-2,-3,9) & (-2,-2,15,-3) \\ (-2,-2,9,-3) & (-2,-2,15,15) & (-2,4,-3,3) \\ (-2,4,3,-3) & (-2,4,3,3) & (1,-2,-3,3) \\ (0,-2,-3,3) & (1,-2,-3,1) & (1,-2,3,-3) \\ (0,-2,3,-3) & (1,-2,1,-3) & (1,-2,3,3) \\ (0,2,-3,-1) & (0,2,-1,-3) & (0,2,-1,-1) \\ (1,0,-3,-1) & (1,0,-1,-3) & (1,0,-1,-1). \end{array} \tag{59}$$

Substitution in (58) gives the source cones as the active sets; every omitted inequality has margin at least one. The recession cone is zero: pairing with v_0, v_1, v_2 first forces the first two coordinates to vanish, and the remaining opposite coordinate pairs force the other two to vanish. Thus these integral vertices give a strictly convex Cartier support function. On $P' = X_{\Sigma'}$, the divisor $A = \sum h_i D_i$ is ample and Cartier. Set

$$\mathcal{L}_0 = A|_{X'_9}, \qquad H = c_1(q^*\mathcal{L}_0). \tag{60}$$

Here X'_9 is the proper transform of a sufficiently general nondegenerate anticanonical member for Δ_9.

For an explicit smooth ambient refinement, replace the heights by

$$a_i = h_i - \frac{\mathbf{1}_{i=3} + \mathbf{1}_{i=4}}{100} + \frac{i^2}{10^6}. \tag{61}$$

The maximal cones of the normal fan of P_a have ray-index sets

$$\begin{gathered}
\{0,1,3,6\},\ \{0,1,3,7\},\ \{0,1,4,5\},\ \{0,1,4,7\},\\
\{0,1,5,6\},\ \{0,2,3,6\},\ \{0,2,3,7\},\ \{0,2,4,5\},\\
\{0,2,4,7\},\ \{0,2,5,6\},\ \{1,3,6,9\},\ \{1,3,7,8\},\\
\{1,3,8,9\},\ \{1,4,5,9\},\ \{1,4,7,8\},\ \{1,4,8,9\},\\
\{1,5,6,9\},\ \{2,3,6,10\},\ \{2,3,7,8\},\ \{2,3,8,10\},\\
\{2,4,5,10\},\ \{2,4,7,8\},\ \{2,4,8,10\},\ \{2,5,6,10\},\\
\{3,6,9,10\},\ \{3,8,9,10\},\ \{4,5,9,10\},\ \{4,8,9,10\},\\
\{5,6,9,10\}.
\end{gathered} \tag{62}$$

Each ray matrix has determinant ± 1, and each cone lies in a cone of Σ' and in a facet of Δ_9. Consequently its toric fourfold T is smooth and projective and the refinement is crepant. At the four source parallelograms

$$q_1 = (3,6,10,8), \quad q_2 = (2,6,3,7), \quad q_3 = (4,5,10,8), \quad q_4 = (2,5,4,7), \tag{63}$$

the refinement uses the diagonal $\{a, c\}$ of $q = (a, b, c, d)$. Thus the anticanonical proper transform $Y \subset T$ is the source's chosen projective small resolution $q : Y \to X'_9$. Extra refinements over facet-interior strata do not change the hypersurface: those strata map to torus-fixed points missed by a nondegenerate member. In particular the additional nonsimplicial coarse cone with rays $\{2, 7, 8, 10\}$ is contained in a single original facet and contributes no fifth node. These model identifications use [W26k, Proposition 3.4].

9.2. **A first-order family with the prescribed ample class.** We state explicitly the additional deformation step. The nodal specializations of CY V used here formulate the classical criterion; no mixed-cone extension of the spectral theorem is asserted.

Lemma 9.1. *Let Z be a connected normal projective nodal threefold over $\mathbb{C}$ with $\omega_Z \simeq \mathcal{O}_Z$ and $H^1(\mathcal{O}_Z) = H^2(\mathcal{O}_Z) = 0$, admitting a projective small crepant resolution. Suppose the exceptional-curve relation space contains a vector with no zero coordinate. For every ample Cartier line bundle $\mathcal{L}_0$ on Z, there is a proper flat projective smoothing over a disc, with trivial relative canonical bundle and a relatively ample line bundle restricting to exactly $\mathcal{L}_0$, such that every node is smoothed to first order.*

Proof. By [W26j, Theorem 2.5], the full-support relation lifts to a global deformation tangent whose image in every local node-smoothing space is nonzero. The common-volume-form

identifications may rescale equation parameters by nonzero factors, which preserve this condition. The deformation base is smooth by [W26j, Theorem 4.5]; its additional degree-six hypothesis is vacuous for nodes.

The projective effectivity construction in Proposition 8.1 applies with this chosen tangent. For completeness, choose N with $\mathcal{L}_0^N$ very ample and $H^1(\mathcal{L}_0^N) = 0$. Across a small Artin extension, the obstruction to lifting this line bundle lies in $H^2(\mathcal{O}_Z)$ and the obstruction to lifting its coordinate sections lies in $H^1(\mathcal{L}_0^N)$. Both vanish. The lifted map remains a closed immersion. Embedded deformations therefore map formally smoothly to abstract deformations, and the Hilbert germ is smooth with a lift of the selected tangent.

Work on a finite-type Hilbert neighbourhood with integral fibres. An étale neighbourhood supplies a section through a smooth point of Z [Stacks, Lemma 37.38.5, Tag 057G]. The relative Picard scheme and its normalized Poincaré bundle then exist as in [Kle05, Theorems 2.5(2), 4.8 and Exercise 4.3]. The Picard morphism is smooth at $[\mathcal{L}_0]$ because $H^2(\mathcal{O}_Z) = 0$ [Kle05, Proposition 5.19]. A further étale-local section through that point extends $\mathcal{L}_0$ itself. Ampleness persists after shrinking. Both changes of base induce tangent isomorphisms. A disc with the chosen tangent in the resulting smooth base retains every nonzero local derivative. The auxiliary power N neither changes the final polarization nor reparametrizes the disc.

The equations $\sum z_j^2 = \tau_i(t)$ with $\tau_i'(0) \neq 0$ make the total space smooth at the nodes. Properness and smoothness away from them permit shrinking so that the total space and every noncentral fibre are smooth. The relative dualizing line bundle restricts to ω_Z. Since $H^1(\omega_Z) = 0$, cohomology and base change lifts a central nowhere-zero generator to a relative section. Its zero locus has proper image missing the origin; after shrinking it is empty. This trivializes the relative canonical bundle. □

Proposition 9.2. *The model X_9', resolution Y and exact polarization (60) admit a smoothing satisfying all geometric hypotheses of Theorem 1.2. Their second Betti numbers are $b_2(Y) = 6$ and $b_2(X_t) = 3$.*

Proof. The earlier construction gives exactly four nodes and no other singularities. Adjunction and crepancy give trivial dualizing sheaves. On the smooth projective toric fourfold T, the anticanonical sequence $0 \to K_T \to \mathcal{O}_T \to \mathcal{O}_Y \to 0$, toric structure-sheaf vanishing and Serre duality give $H^1(\mathcal{O}_Y) = H^2(\mathcal{O}_Y) = 0$. Rationality of the nodes transfers the vanishings to X_9'. The source's full integral curve-relation lattice is

$$\left\{ \ell \in \mathbb{Z}^4 : \sum_i \ell_i [C_i] = 0 \right\} = \mathbb{Z}(1, -1, -1, 1) \tag{64}$$

by [W26k, Proposition 3.5 and Section 11.1]. Lemma 9.1 therefore supplies the exact polarized first-order smoothing and trivial relative canonical bundle.

For simple connectivity, apply [BK06, Corollary 1.9] to the nondegenerate hypersurface and the smooth projective crepant fan above. The relevant lattice contains all original vertices, in particular the basis v_0, v_1, v_3, v_4 of $\mathbb{Z}^4$. Its quotient, hence $\pi_1(Y)$, is zero. Van Kampen across the simply connected links and fillings gives $\pi_1(X_t) = 0$.

The source divisors span $H^2(Y; \mathbb{Q})$ [W26k, Lemma 3.3]. There are four independent character relations, and $D_9|_Y = 0$ because the facet-interior divisor misses the hypersurface. Thus $b_2(Y) \leq 6$. The nonsingular intersection matrix (66) below proves that six indicated classes are independent. The exceptional span has dimension three by (64), so the attachment sequence gives $b_2(X_t) = 6 - 3 = 3$. □

This constructs a suitable smoothing of the exact earlier central model. CY VI's Theorem 9.25 supplies a proper analytic smoothing with projective fibres; it does not by itself assert the precise relative polarization or first-order condition used here. We do not identify the chosen disc with that theorem's arbitrary smoothing arc, or its metrics with those of a birational mirror model.

9.3. **The three-dimensional harmonic quotient.** Use the rational basis of integral divisor classes

$$(D_2, D_5, D_6, D_7, D_8, D_{10}).$$

No assertion that this is an integral lattice basis is needed. The character relations, $D_9|_Y = 0$ and (58) give

$$H = (6, 18, 18, -12, -10, 8), \qquad H^3 = 8712. \tag{65}$$

The polarized intersection matrix is

$$B = \left(\int_Y D_i D_j H\right) = \begin{pmatrix} -18 & 6 & 6 & 6 & 0 & 0 \\ 6 & 0 & 14 & 0 & 0 & 2 \\ 6 & 14 & 0 & 0 & 0 & 2 \\ 6 & 0 & 0 & -22 & 12 & 0 \\ 0 & 0 & 0 & 12 & -16 & 2 \\ 0 & 2 & 2 & 0 & 2 & -6 \end{pmatrix}, \qquad \det B = -5064192. \tag{66}$$

These are intersection products in the toric ring with character relations $\sum_i \langle m, v_i\rangle D_i = 0$ and the Stanley–Reisner relations of (62). Every maximal cone product integrates to one, and $[Y] = \sum_i D_i$. These relations, together with the displayed rays and cones, determine every entry of (66) and the volume in (65).

For $q = (a, b, c, d)$ with the ac diagonal, the integral source circuit is $e_b + e_d - e_a - e_c$. In the chosen basis the four circuits give

$$C = \begin{pmatrix} 0 & 0 & 1 & 0 & 1 & -1 \\ -1 & 0 & 1 & 1 & 0 & 0 \\ 0 & 1 & 0 & 0 & 1 & -1 \\ -1 & 1 & 0 & 1 & 0 & 0 \end{pmatrix}. \tag{67}$$

It has rank three, with the full surviving kernel spanned by

$$k_1 = D_2 + D_7, \quad k_2 = D_5 + D_6 - D_7 + D_{10}, \quad k_3 = D_8 + D_{10}. \tag{68}$$

In particular $H = 6k_1 + 18k_2 - 10k_3$. Choose quotient lifts (D_2, D_5, D_6). The energy and the full Schur complement are

$$E = \begin{pmatrix} 2 & -1 & -1 \\ -1 & 2 & 0 \\ -1 & 0 & 2 \end{pmatrix}, \qquad M = \frac{1}{15}\begin{pmatrix} 177 & -79 & -79 \\ -79 & 158 & -52 \\ -79 & -52 & 158 \end{pmatrix}. \tag{69}$$

The leading principal minors of E are $2, 3, 4$; those of M are $59/5, 869/9, 17584/45$. Both forms are positive definite. Direct expansion gives

$$\frac{\det(E - \mu M)}{-\det M} = \left(\mu - \frac{1}{7}\right)\left(\mu^2 - \frac{375}{628}\mu + \frac{45}{628}\right). \tag{70}$$

The ordered roots are $1/7$ and $(375 \mp 3\sqrt{3065})/1256$. They are positive and distinct. Proposition 9.2 verifies the geometric assumptions, and Theorem 1.2 now proves Corollary 1.5, including its multiplicities, type and gaps.

The earlier vector $(1,-1,-1,1)$ is the kernel of the curve-class map A and provides a full-support smoothing tangent. Its annihilator is the three-dimensional period image of $C = A^*$ and counts the small modes. The new coefficients additionally use the global form (66) and projection off all three classes in (68). The mirror equality of relation lattices in CY VI is not used to calculate that norm. Thus the earlier construction and deformation theorems are actual premises of this application, while the quantitative spectral conclusion follows from the analysis of the present paper.

10. Further directions

The restriction map connects three parts of the transition: its kernel is the surviving second cohomology, its image gives the integral relations among cohomological vanishing cycles, and its periods give the logarithmic spectral energy. The polarized harmonic quotient supplies the mass. The split-quintic calculation makes the associated integral rings and attachments explicit; the toric examples show how a higher-dimensional quotient produces several distinct coefficients.

For an actual base change $s = t^k$ of a family satisfying the first-order hypotheses, the polarized fibre at t is exactly the old fibre at $s = t^k$. Ricci-flat uniqueness therefore gives

$$\lambda^+_{2,j}(t) = \frac{2\pi\mu_j}{k|\log t|}(1 + o(1)).$$

Counts, gaps and type are inherited, as is the rank-one one-sided remainder for fixed k. Equal local smoothing orders alone do not imply that an arbitrary global family descends through this base change. Different node orders or ratios approaching zero require metric estimates uniform on the enlarged parameter domain; the separate cap capacities do not supply them.

The proof gives a leading coefficient for each fixed family. A two-sided next-order expansion would require quantitative control of the limiting spectral subspaces and the full harmonic projection. Uniformity as the quotient spectral matrix itself degenerates is a further problem. The small two-form modes have exact three-form partners under d, and Hodge duality gives related modes in degree four; the argument does not exhaust the middle-degree small spectrum.

The tangent results in Appendix B concern connections induced by the Ricci-flat geometry. Extending the analysis to a separately specified bundle would require compatible deformation and stability data, uniform bundle inverses, and control of possible curvature concentration. Such bundle spectra and HYM flows are separate from the Hodge theorem proved here.

Appendix A. Integral topology of the marked split quintic

Choose independent linear forms l_1, l_2 and quartics q_1, q_2 on $\mathbb{P}^4$. Put $\Pi = \{l_1 = l_2 = 0\}$ and assume that the restricted quartics meet transversely in sixteen points, that $X_0 = \{l_1q_2 - l_2q_1 = 0\}$ is smooth away from Π, and that

$$Y = \{ul_1 + vl_2 = 0,\ uq_1 + vq_2 = 0\} \subset \mathbb{P}^4 \times \mathbb{P}^1$$

is smooth. These are nonempty open conditions. Choose a quintic G nonzero at the sixteen nodes and let $X_t = \{l_1q_2 - l_2q_1 + tG = 0\}$, for sufficiently small t. The projective construction, the hyperplane markings H, L, and the chosen smoothing ray are the global inputs. The complement ring and attachment maps below are calculated, rather than assumed.

Let U be the common compact exterior, $q : Y \to X_0$ the projection, and $s_t : X_t \to X_0$ the continuous Milnor collapse along the ray. Mark the degree-two link generators by the

exceptional rulings. Write

$$\Lambda = \mathbb{Z}^{16}, \qquad \mathbf{1} = (1, \dots, 1), \qquad \Lambda_0 = \ker(\Sigma : \Lambda \to \mathbb{Z}), \qquad W = \Lambda/\mathbb{Z}\mathbf{1}.$$

Here Σ is coordinate summation. Let J_{15} be the 15×15 all-ones matrix.

Theorem A.1 (The marked split-quintic transition)**.** *The preceding construction has the following compatible properties.*

(1) *The smooth threefolds Y and X_t are simply connected and have torsion-free integral cohomology, with the full rings and Chern classes in Table 1. Their third Betti numbers are* 174 *and* 204*; their degree-three products are unimodular alternating pairings times the point class. In both cases $H^0 = H^6 = \mathbb{Z}$ and $H^1 = H^5 = 0$. The integral groups of U and X_0 are free, with Betti vectors*

$$(1, 0, 2, 189, 1, 15, 0), \qquad (1, 0, 1, 189, 2, 0, 1),$$

respectively. The central ring and both specialization pullbacks are given by (79) and the degree-three description in Section A.3; in particular $\operatorname{im} s_t^* = \ker(T^* - I)$ in integral degree-three cohomology, where T is monodromy about a positive simple loop.

(2) *The degree-two boundary restriction is $r(hH + \ell L) = \ell\mathbf{1}$. It occurs in the integral exact sequence*

$$0 \longrightarrow \mathbb{Z}H \longrightarrow \mathbb{Z}H \oplus \mathbb{Z}L \xrightarrow{r} \Lambda \xrightarrow{\partial} H^3(X_t; \mathbb{Z}) \longrightarrow H^3(U; \mathbb{Z}) \longrightarrow 0. \tag{71}$$

The classes ∂e_i are the left Poincaré duals of the vanishing spheres. They generate the primitive isotropic lattice W, with the single relation $\sum_i \partial e_i = 0$. Under Poincaré duality the nonzero monodromy block is $\pm(I_{15} + J_{15})$, whose Smith form is $\operatorname{diag}(1^{14}, 16)$. Thus the monodromy image has index sixteen in the primitive vanishing lattice.

(3) *Use the actual Ricci-flat class $H + s^2 L$ on Y and class H on X_t, where $s, t > 0$ are independent branch parameters. Both branches converge to the same metric completion (X_0, d_{g_0}), homeomorphic to X_0, with the local rescaled resolved-conifold and Stenzel limits. Every exceptional area on Y equals s^2. The same class L has period one at every link, and its central primitive harmonic representative α satisfies*

$$[\alpha] = L - \tfrac{4}{5}H, \qquad \|\alpha\|^2_{L^2(g_0)} = \tfrac{16}{5}, \qquad \lim_{t \downarrow 0} |\log t| \lambda_2^+(t) = \frac{2\pi \|r(L)\|^2}{16/5} = 10\pi. \tag{72}$$

The eigenvalue has the simplicity, type, gap and one-sided upper-error properties of Theorem 1.1. The resolution and smoothing limits require no relation between s and t.

(4) *The tangent bundles on the smooth spaces are stable and their Ricci-flat tangent connections are Hermitian–Einstein. The relative tangent sheaf of the smoothing is flat, with central fibre E_0 and*

$$0 \longrightarrow E_0 \longrightarrow T_{X_0} \longrightarrow \bigoplus_{i=1}^{16} \mathbb{C}_{p_i} \longrightarrow 0, \qquad E_0^{**} = q_* T_Y = T_{X_0}, \qquad R^j q_* T_Y = 0 \ (j > 0).$$

These extensions agree on the regular locus. Their actual tangent connections have the local limits and curvature-energy bounds in Proposition B.2.

TABLE 1. Integral cohomology and characteristic data of the compact fibres.

	Y	X_t
H^2	$\mathbb{Z}H \oplus \mathbb{Z}L$	$\mathbb{Z}H$
H^4	$\mathbb{Z}A \oplus \mathbb{Z}B$	$\mathbb{Z}a_4$
Degree-two products	$H^2 = 5A + 4B$, $HL = 4A$, $L^2 = 0$	$H^2 = 5a_4$
Top products	$HA = LB = [\mathrm{pt}]$, $HB = LA = 0$	$Ha_4 = [\mathrm{pt}]$
$\operatorname{rank} H^3$	174	204
c_2	$50A + 24B$	$50a_4$
Euler characteristic	-168	-200

A.1. **The full integral rings.** Let H, L be the hyperplane classes of $\mathbb{P}^4, \mathbb{P}^1$ restricted to Y. Lefschetz gives $H_2(Y;\mathbb{Z}) = \mathbb{Z}^2$ and $H_2(X_t;\mathbb{Z}) = \mathbb{Z}$. Together with $H_1 = 0$, universal coefficients and integral Poincaré duality imply torsion-freeness in every degree. The cubic intersections are (51). Take $A, B \in H^4(Y;\mathbb{Z})$ dual to H, L, and let $a_4 \in H^4(X_t;\mathbb{Z})$ satisfy $\int_{X_t} Ha_4 = 1$. Then:

To verify the characteristic numbers directly, use

$$c(TY) = \frac{(1+H)^5(1+L)^2}{(1+H+L)(1+4H+L)}.$$

This gives $c_2(Y) = 6H^2 + 5HL$ and $c_3(Y) = -20H^3 - 17H^2L$. For a quintic, $c_2 = 10H^2$ and $c_3 = -40H^3$. Since $H^1 = H^5 = 0$, the Euler numbers give the displayed third Betti numbers. The degree-three product on each compact fibre is its unimodular alternating pairing times the point class. With $H^0 = H^6 = \mathbb{Z}$, these statements specify the full rings up to symplectic basis choices. The product subgroup generated by H^2 and HL in $H^4(Y)$ has index $|\det \left(\begin{smallmatrix} 5 & 4 \\ 4 & 0 \end{smallmatrix}\right)| = 16$; it is not an integral basis.

Every exceptional curve has degrees $(H \cdot C_i, L \cdot C_i) = (0, 1)$. They therefore represent the same primitive homology class c_Y, with Poincaré dual B. Primitivity follows from the period of L.

A.2. **The common exterior and all cohomological attachments.** Remove disjoint tubular neighborhoods of the exceptional curves, obtaining a compact exterior U. Its boundary components $\mathcal{L}_i$ are $S^2 \times S^3$. Define

$$\Lambda = \mathbb{Z}^{16}, \quad \mathbf{1} = (1, \dots, 1), \quad \Lambda_0 = \ker(\Sigma : \Lambda \to \mathbb{Z}), \quad W = \Lambda/\mathbb{Z}\mathbf{1}.$$

The homological Thom sequence uses $H_j(Y, U) = H_{j-4}(\coprod C_i)$. The maps $H_4(Y) \to \Lambda$ and $H_6(Y) \to \Lambda$ have primitive diagonal image: the former is exceptional intersection, witnessed by $L \cdot C_i = 1$, and the latter is the oriented diagonal fundamental class. Thus U has free integral homology and cohomology, with Betti vector

$$(1, 0, 2, 189, 1, 15, 0), \tag{73}$$

and exact sequences

$$0 \to W \to H_3(U) \to H_3(Y) \to 0, \qquad 0 \to H_3(U) \to H_3(X_t) \to \Lambda_0 \to 0. \tag{74}$$

The second comes from the Milnor fillings, whose cores are three-spheres. The vanishing spheres span W, with relation $\delta_1 + \cdots + \delta_{16} = 0$. Both successive quotients in (74) are free; hence W embeds primitively in $H_3(X_t)$, and is isotropic because the spheres are disjoint.

In cohomology the Thom sequence kills exactly $\mathbb{Z}B$ in degree four. Writing $\bar{A} = A|_U$, the even products are

$$H^2 = 5\bar{A}, \qquad HL = 4\bar{A}, \qquad L^2 = 0.$$

Orient each link as a boundary component of U and choose generators $\alpha_i \in H^2(\mathcal{L}_i)$ and $\beta_i \in H^3(\mathcal{L}_i)$ with $L|_{\mathcal{L}_i} = \alpha_i$ and $\int \alpha_i\beta_i = 1$. The symbol β_i here has degree three; the harmonic degree-two form remains ϑ_i. The degree-three restriction is

$$0 \to H^3(Y) \to H^3(U) \xrightarrow{r_U} \Lambda_0 \to 0, \tag{75}$$

and degree-five restriction is the inclusion $H^5(U) = \Lambda_0 \hookrightarrow \Lambda$. Boundary multiplicativity and that injectivity imply, for $y \in H^3(U)$,

$$Hy = 0, \qquad Ly = r_U(y) \in \Lambda_0 = H^5(U).$$

All degree-six products vanish. These formulas give the entire exterior ring, without supplying an unspecified multiplication as additional input.

For completeness, use the Mayer–Vietoris difference $j_U^* - j_F^*$. Let x_i be the degree-two generator of the resolution filling and z_i the degree-three generator of the smoothing filling, restricting to α_i, β_i. The nonzero difference maps are:

Degree	Resolution	Smoothing
0	$c_0\mathbf{1} - (c_i)$	$c_0\mathbf{1} - (c_i)$
2	$\ell\mathbf{1} - (x_i)$	$\ell\mathbf{1}$
3	$r_U(y)$	$r_U(y) - (z_i)$
5	$\Lambda_0 \hookrightarrow \Lambda$	$\Lambda_0 \hookrightarrow \Lambda$

Here $hH + \ell L$ is the exterior degree-two class, c_0 is an exterior constant, and (c_i) are filling constants. The nonzero connecting maps are

$$\begin{aligned} \partial_Y^3(v) &= (\Sigma v)B, & \partial_Y^5(v) &= (\Sigma v)[\mathrm{pt}], \\ \partial_{X_t}^2 : \Lambda \twoheadrightarrow W \hookrightarrow H^3(X_t), & & \partial_{X_t}^5(v) &= (\Sigma v)[\mathrm{pt}]. \end{aligned} \tag{76}$$

All other connecting maps vanish. The signs follow by representing the connecting class on an outward-U collar by $d\chi_F \wedge \eta$, where χ_F increases toward the filling. Pairing the degree-three connecting class of β_i with H, L gives zero and one respectively, hence $+B$. The same collar integral gives the top-degree sum maps.

Restriction from Y preserves H, L, sends (A, B) to $(\bar{A}, 0)$, and identifies $H^3(Y)$ with $\ker r_U$. Restriction from X_t sends (H, a_4) to $(H, \bar{A})$ and surjects $H^3(X_t) \to H^3(U)$ with kernel the cohomological copy of W in (76). Restriction to each Milnor filling is the corresponding component of $r_U(y)$. These data specify the full integral cohomological attachment diagram; no homotopy classification of the attaching maps is asserted.

A.3. **Monodromy and specialization.** Use left Poincaré duals in odd degree, characterized by $\int \mathrm{PD}_L(\delta) \wedge \eta = \int_\delta \eta$, and orient δ_i so that $\mathrm{PD}_L(\delta_i) = \partial_{X_t}^2(\alpha_i)$. For a positive simple loop, the Picard–Lefschetz transformation has $N = T - I$, $N^2 = 0$, and, up to the common intersection-sign convention,

$$N\gamma = \pm \sum_{i=1}^{16} \langle \gamma, \delta_i \rangle \delta_i. \tag{77}$$

Equivalently it factors through $H_3(X_t) \to \Lambda_0 \hookrightarrow \Lambda \twoheadrightarrow W$. In the bases $e_i - e_{16}$ of Λ_0 and $\delta_1, \dots, \delta_{15}$ of W, the nonzero block is $\pm(I_{15} + J_{15})$, where J_{15} is the all-ones matrix. Its Smith form is $\mathrm{diag}(1^{14}, 16)$: subtracting the first row from the others gives fourteen unit pivots, and

the remaining determinant is sixteen. Thus the monodromy image has index sixteen in the primitive vanishing lattice. This relation-matrix description is consistent with the general conifold monodromy calculation of [LLW18].

The contraction $q : Y \to X_0$ collapses each exceptional sphere to its node. Along the chosen smoothing ray, the Milnor collapse $s_t : X_t \to X_0$ collapses each vanishing sphere. A disk cotangent bundle of S^3 modulo its zero section is the cone on its sphere bundle; the local collapses glue to the common exterior identification. Both maps have degree one. Collapsing the sixteen cores to their distinct nodes identifies the corresponding relative pairs with $(X_0, \{p_i\})$. In degrees at least two its relative groups are $H_j(X_0)$. The pair sequences give

$$\begin{aligned} &0 \to H_3(Y) \xrightarrow{q_*} H_3(X_0) \to \Lambda_0 \to 0, \\ &0 \to W \to H_3(X_t) \xrightarrow{s_{t*}} H_3(X_0) \to 0. \end{aligned} \tag{78}$$

In degree two, q_* is the quotient by $\mathbb{Z}c_Y$ and s_{t*} is an isomorphism. In degree four, q_* is an isomorphism and s_{t*} is a primitive injection with free rank-one cokernel. To see the latter, the full relative sequence is

$$0 \to H_4(X_t) \to H_4(X_0) \to \Lambda \xrightarrow{e_i \mapsto \delta_i} H_3(X_t) \to H_3(X_0) \to 0,$$

and the middle kernel is $\mathbb{Z}\mathbf{1}$. All central groups are free, with Betti vector $(1, 0, 1, 189, 2, 0, 1)$. The maps are isomorphisms in degrees zero and six; degrees one and five vanish.

The first sequence of (74) maps to the second sequence of (78), with identity on W, exterior inclusion in the middle, and q_* on the right. Both vertical cokernels are canonically Λ_0. This is naturality of the actual collapse maps, rather than an arbitrary splitting of free groups.

Let $h \in H^2(X_0)$ be the central polarization and choose $A_0, B_0 \in H^4(X_0)$ by $q^*(h, A_0, B_0) = (H, A, B)$. The central even products and smoothing pullback are

$$h^2 = 5A_0 + 4B_0, \quad hA_0 = [\mathrm{pt}], \quad hB_0 = 0, \qquad s_t^*(h, A_0, B_0) = (H, a_4, 0). \tag{79}$$

Indeed pairing the degree-four pullbacks with H, and using degree one, fixes their coefficients integrally. Dually the image of s_{t*} in degree four is $\mathbb{Z}q_* \operatorname{PD}_Y(H)$. In degree three q^* is surjective with primitive kernel $\Lambda_0^* \simeq W$, and s_t^* is a primitive injection onto the annihilator of the vanishing homology lattice. Since $\operatorname{im} N$ has finite index in W, an integer-valued homomorphism annihilates $\operatorname{im} N$ exactly when it annihilates W. Thus

$$\operatorname{im} s_t^* = \ker(T^* - I) \subset H^3(X_t; \mathbb{Z}).$$

The central degree-three cup pairing factors through q^* and the unimodular pairing of Y, by naturality and degree one. Its radical is $\ker q^*$; the Milnor pullback takes that radical to the primitive vanishing cohomology lattice by naturality of the connecting map. These formulas specify the central ring, including its degenerate middle pairing, without assuming ordinary Poincaré duality for X_0.

Appendix B. Tangent sheaves and Hermitian–Einstein limits

The tangent bundles provide compatible nontrivial holomorphic data on the same transition. Their Hermitian–Einstein connections are intrinsic to the Ricci-flat metrics, but their algebraic extensions across the central fibre require a separate calculation. We make that distinction explicit.

B.1. **The flat limit and its reflexive hull.** For the family (50), put

$$\mathcal{A} = T_{\mathbb{P}^4}|_{\mathcal{X}}, \qquad \mathcal{B} = \mathcal{O}_{\mathcal{X}}(5), \qquad \mathcal{K} = \ker(\mathcal{A} \xrightarrow{d_{\mathbb{P}^4}(F_0+tG)} \mathcal{B}).$$

Thus $\mathcal{K} = T_{\mathcal{X}/\Delta}$, and set $E_0 = \mathcal{K}|_{X_0}$. Write $Q = \bigoplus_{i=1}^{16} \mathbb{C}_{p_i}$.

Proposition B.1. *The sheaf $\mathcal{K}$ is flat over the disc and restricts to T_{X_t} for $t \neq 0$. There are identities and an exact sequence*

(80)
$$0 \to E_0 \to T_{X_0} \to Q \to 0, \qquad E_0^{**} = T_{X_0}, \qquad q_* T_Y = T_{X_0}, \qquad R^j q_* T_Y = 0 \quad (j > 0).$$

At each node T_{X_0} is reflexive of depth two, whereas E_0 has depth one and is not reflexive. On each exceptional curve,

$$T_Y|_{C_i} = \mathcal{O}(2) \oplus \mathcal{O}(-1)^{\oplus 2}. \tag{81}$$

Proof. Let $\mathcal{I}$ and $\mathcal{Q}$ be the image and cokernel of the relative Jacobian map. Locally the total space is smooth and the projection has equation $\sum z_j^2 = t$. Hence $\mathcal{Q}$ is a length-one skyscraper at each node, annihilated by t, and vanishes elsewhere near the central fibre. Both $\mathcal{I} \subset \mathcal{B}$ and $\mathcal{K} \subset \mathcal{A}$ have no base-parameter torsion, so they are flat over the smooth one-dimensional base. Tensor the sequences

$$0 \to \mathcal{K} \to \mathcal{A} \to \mathcal{I} \to 0, \qquad 0 \to \mathcal{I} \to \mathcal{B} \to \mathcal{Q} \to 0$$

with the central fibre. The first stays left exact, and the second gives $\ker(\mathcal{I}_0 \to \mathcal{B}_0) = \operatorname{Tor}_1^{\mathcal{O}_\Delta}(\mathcal{Q}, \mathbb{C}) = Q$. The kernel of $\mathcal{A}_0 \to \mathcal{B}_0$ is T_{X_0}, so the snake lemma gives the first sequence in (80).

At a node let $R = \mathbb{C}\{z_1, \dots, z_4\}/(\sum z_j^2)$ and let $\mathfrak{m}$ be its maximal ideal. The tangent module is the kernel of $R^4 \to \mathfrak{m}$. The sequence $0 \to \mathfrak{m} \to R \to \mathbb{C} \to 0$ gives depth one for $\mathfrak{m}$, and the kernel consequently has depth two. It is torsion-free and locally free off the node, hence reflexive on this normal threefold. The sequence with nonzero length-one quotient then gives depth one for E_0. Both sheaves agree off codimension three, so their reflexive hulls agree.

The exceptional locus in the smooth Y has complex codimension two. Holomorphic sections of its tangent bundle extend across those curves. On the common regular locus this identifies $q_* T_Y$ with the reflexive extension T_{X_0} [Stacks, Tags 0EBJ and 0AY6]. For the higher direct images, the tangent-normal sequence over C_i splits because $H^1(\mathbb{P}^1, \mathcal{O}(3)) = 0$, giving (81). If I is the exceptional ideal, $I^n/I^{n+1} = \mathcal{O}(n)^{\oplus(n+1)}$. Tensoring with the tangent restriction gives degrees $n+2$ and $n-1$, whose first cohomology vanishes for every $n \geq 0$. Induction through the thickening sequences therefore gives $H^1(T_Y/I^{n+1}T_Y) = 0$. In an incidence chart the node coordinates are the entries of $\left(\begin{smallmatrix} a & b \\ ua & ub \end{smallmatrix}\right)$; the pulled-back node ideal is $(a, b) = I$. These are exactly the thickenings required by formal functions, not merely an equivalent informal neighborhood model. Formal functions and faithful flatness of completion give $R^1 q_* T_Y = 0$ [Stacks, Tag 02OC]. The fibres have dimension at most one, so all higher direct images vanish as well. □

The exceptional splitting rules out locally free descent across the contracted point. In particular the resolution direct image is the reflexive hull, not the flat central fibre E_0. Compatibility is through the common regular tangent bundle and the specified coherent extensions. The exterior identifications on the smoothing side are differentiable, not generally biholomorphic. No single flat family containing both the resolution and smoothing branches is asserted.

Every smooth deformation has its intrinsic relative tangent bundle. Thus this bundle choice introduces no additional existence obstruction on smooth base deformations. The singular flatness calculation above uses the actual one-dimensional family with nonzero linear local smoothing terms. It is not a flatness theorem over an arbitrary higher-dimensional singular deformation base. As a consistency check, proper flatness, (80) and the direct images imply the holomorphic Euler-characteristic identity $\chi(Y,T_Y)-\chi(X_t,T_{X_t})=16$, distinct from the topological Euler-characteristic difference 32.

B.2. **Actual metrics and controlled local curvature.** On the resolution use the Kähler class $H+\epsilon^2 L$, the restriction of a positive product class, and on the smoothing use H with $t=\epsilon^3$. Denote either branch by Z_ϵ, with its actual Ricci-flat metric g_ϵ.

Proposition B.2. *The nontrivial tangent bundles T_Y and T_{X_t} are stable for these classes. Their Ricci-flat tangent connections are irreducible zero-slope Hermitian–Einstein connections. Both branches converge to the same metric completion (X_0,d_{g_0}), homeomorphic to X_0, smoothly on fixed regular compact subsets. Near each node, after rescaling by ϵ^{-2}, the metrics and tangent connections converge with all derivatives to the normalized nonflat resolved-conifold or Stenzel model.*

For each node there is a collar radius r, extended over the cap with $r+\epsilon\asymp\epsilon$ there, and fixed constants such that

$$
\begin{aligned}
&|\nabla^j \operatorname{Rm}(g_\epsilon)|\le C_j(r+\epsilon)^{-2-j}, \qquad j\ge 0,\\
&\operatorname{Vol}\{r<\rho\}\le C\rho^6, \qquad \int_{\{r<\rho\}}|F_{T_{Z_\epsilon}}|^2dV\le C\rho^2, \qquad \epsilon\le\rho\le\rho_0,\\
&d_{GH}((Z_\epsilon,g_\epsilon),(X_0,d_{g_0}))\le C\epsilon^\theta \quad \text{for some } \theta>0.
\end{aligned}
\tag{82}
$$

All tensor norms, derivatives and volumes in this display use the actual metric. In particular the unscaled tangent curvature-energy measure has no atom at a node.

Proof. Existence of the smooth Ricci-flat metrics is Yau's theorem [Yau78]. These compact simply connected Kähler threefolds have full SU(3) holonomy. A nontrivial de Rham decomposition would have a one-complex-dimensional Ricci-flat factor, whose compact simply connected cover cannot exist; the irreducible hyperkähler alternative has even complex dimension. Ricci-flatness makes the tangent connection Hermitian–Einstein of zero slope; irreducibility and the Hermitian–Einstein stability implication give stability [Kob87, Theorem 5.8.3]. These bundles are nontrivial since their second Chern pairing with H is 50.

The common metric completion and transition convergence are given by [RZ11, Theorem 1.1], and its identification with the nodal variety by [Son15, Theorem 1.1 and Corollary 1.2]. For the local estimates and quantitative distance bound we use the finer expansions. The actual smoothing expansion was verified in Proposition 2.2. On the resolution all local areas are $\int_{C_i}(H+\epsilon^2L)=\epsilon^2$. Thus the local AC Kähler classes lie in the restriction image of the single global class L, exactly Assumption R of Benabida's Theorem B [Ben26]; his Example 1.5 also treats this split quintic. For completeness, the finite-node scalar estimate in Proposition 2.2 applies equally to the disjoint union of resolved-conifold caps. The low central holomorphic germs pull back through each small resolution, so their local $\partial\bar\partial$ vanishes there. The formal tensor orders and uniform corner extraction therefore apply on this branch as well. In [Ben26, §5.3.1], choose the initial gluing with the specified global class L under Assumption R. Its class is exactly $H+\epsilon^2L$ in our normalization, and all later corrections are $\partial\bar\partial$-exact. Uniqueness identifies the resulting metric with the prescribed resolution metric. The face restrictions give the claimed central and rescaled local limits, with all derivatives.

For tangent connections these follow by differentiation of the actual metrics and complex structures.

The positive face metrics and the bounded differentiated coefficients in conic frames give uniform cone comparison on annuli and scaled AC comparison on the core. Every physical derivative costs $(r+\epsilon)^{-1}$; two derivatives give curvature. This proves the first line of (82) and core volume $O(\epsilon^6)$. Integrating the annular volume comparison gives the displayed $C\rho^6$ bound. On a dyadic annulus of radius $s \geq \epsilon$, the tangent curvature energy is bounded by $Cs^{-4}s^6 = Cs^2$. Summing to the core, whose energy is $O(\epsilon^2)$, gives $C\rho^2$. The central cone comparison adds one point in the metric completion at each node and identifies its topology with the algebraic cone, so the completion is homeomorphic to X_0.

We give the global distance estimate explicitly. Each filled region $\{r < \rho\}$ has intrinsic diameter at most $C\rho$: radial annular lengths sum to $O(\rho)$, fixed-radius links have diameter $O(\rho)$, and the core has diameter $O(\epsilon)$. Collapsing the sixteen regions to points changes Gromov–Hausdorff distance by at most $C\rho$. Indeed a shortest quotient chain need visit each collapsed point at most once, and each jump can be replaced by a path of its intrinsic diameter. The same applies to the central space. On the common exterior the actual tensor comparison, and its resolution analogue from Theorem B, give relative error $O((\epsilon/\rho)^{a_0})$. The resulting quotient spaces have uniformly bounded diameter and are bilipschitz with factor $1 + O((\epsilon/\rho)^{a_0})$. Consequently

$$d_{GH}(Z_\epsilon, X_0) \leq C\big(\rho + (\epsilon/\rho)^{a_0}\big).$$

Taking $\rho = \epsilon^{a_0/(1+a_0)}$ proves the last line of (82), with a possibly smaller common exponent for the two branches. □

The energy statement allows nonflat rescaled limits and unbounded pointwise curvature. It supplies neither uniform estimates for arbitrary gauge bundles nor a theorem for their HYM flows. Its role here is to identify the actual tangent geometry compatible with the explicitly calculated transition.

B.3. **Proof of the integrated theorem.**

Proof of Theorem A.1. Proposition 7.1 establishes the genuine projective family, the chosen resolution and the smooth fibres. The complete intersection calculation in Appendix A gives the torsion-free smooth rings, their Chern classes and pairings. The Thom and Mayer–Vietoris calculations there determine the exterior ring and all cohomological attachments. The contraction and Milnor collapse give the central ring and specialization maps by the relative pair sequences, with the naturality described in Section A.3. This proves part (1).

Use the collar-compatible trivialization of Proposition 2.2 as the exterior identification in the attachment calculation. Its horizontal lift agrees with the prescribed node collars and is defined over the fixed ray. Two such extensions away from the collars are isotopic along that ray, so they induce the same marked cohomology maps. Thus the analytic and topological link generators can be chosen identically. Proposition 3.3 gives (71), since the hyperplane class H is primitive on the smooth quintic and $r(L) = \mathbf{1}$. The freeness of $H^3(U;\mathbb{Z})$ makes its vanishing sublattice primitive. The disjoint vanishing spheres make it isotropic. The calculated factorization $\Lambda_0 \hookrightarrow \Lambda \twoheadrightarrow W$ gives the displayed Picard–Lefschetz block and its Smith form, proving part (2).

For part (3), the same integral class L has $L \cdot C_i = 1$ and cubic data $(H^3, LH^2, L^2H) = (5, 4, 0)$. It determines the exceptional areas s^2 in class $H + s^2L$. On the central regular locus, Proposition 3.1 represents $L - 4H/5$ by the primitive harmonic form of squared norm

16/5. Identity (31) identifies its sixteen boundary coefficients with the same $r(L) = \mathbf{1}$. The spectral theorem therefore gives 10π, with its stated count, type and error. Proposition B.2 supplies the actual metric limits on the two branches. Apply its resolution assertion with $\epsilon = s$ and its smoothing assertion with $\epsilon = t^{1/3}$ separately; no ratio of s and t is used.

Finally, Proposition B.1 identifies the flat central tangent sheaf and its distinct reflexive hull, and Proposition B.2 proves the tangent stability and actual connection limits. This proves part (4). These statements use the same central regular tangent bundle; they do not identify the flat central sheaf with the resolution direct image, or assert a flat family containing both branches. The restriction, primitive norm and actual collar identifications above provide the compatibility between the topological and analytic assertions. □

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